Anticoagulants beyond coagulation: a narrative review of cardiovascular effects

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Abstract

Anticoagulants are widely used to prevent and treat thrombotic disorders, and continuous advances have refined their efficacy and safety. Beyond their established anticoagulant function, growing evidence indicates that some agents exert additional cardiovascular effects through pleiotropic mechanisms.

In this narrative review, we examine both the intended and potential unintended effects of anticoagulants, with particular attention to their actions beyond coagulation. We discuss their influence on vascular pathology and cardiac function, highlighting distinctions between different anticoagulant classes. A deeper understanding of these pleiotropic effects may help guide future research, improve long-term outcomes, and support more targeted therapeutic strategies for patients most likely to benefit.

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1. Introduction

Anticoagulants inhibit the coagulation cascade to prevent or treat thrombosis. Historically, two main classes emerged in the early 20th century. The first comprised unfractionated heparin (UFH), glycosaminoglycans of variable chain length purified from animal tissue, which accelerate antithrombin (AT)-dependent inactivation of serine proteases, including FXa and thrombin. The next generation included low molecular weight heparins (LMWHs), less heterogeneous and enriched in AT-binding capacity, and the synthetic pentasaccharide fondaparinux.[1]

The second class of oral anticoagulants (OACs) consisted of vitamin K antagonists (VKAs), based on the discovery that coumarins—derived from dicoumarol in spoiled sweet clover—inhibit vitamin K–dependent γ-carboxylation of coagulation proteins.[2] Over half a century later, direct OACs (DOACs) were developed as small molecules targeting FXa or thrombin. More recently, FXIa inhibitors have emerged as a novel class with potential safety advantages. In addition, RNA silencing therapies and monoclonal antibodies, mostly targeting FXI(a), are in (pre)clinical testing.[3]

While anticoagulants aim to prevent thrombosis, their principal unintended effect is bleeding. Both efficacy (thrombus inhibition) and safety (bleeding risk) depend on anticoagulation intensity. Beyond fibrin inhibition, anticoagulants can also modulate fibrinolysis, platelets, and endothelial function, contributing to both antithrombotic and bleeding-prone actions, as well as pleiotropic effects, including anti-inflammatory properties. This review examines the intended and unintended cardiovascular effects of current anticoagulants.

2. Pharmacological properties of anticoagulants

2.1 Parenteral anticoagulants: (low molecular weight) heparins and direct thrombin inhibitors

UFH and LMWH act by amplifying the rate of coagulation protease inactivation by the natural serine protease inhibitor AT.[4] Binding of UFH to AT results in a conformational change in the latter, thereby rendering it a rapid and irreversible inhibitor of FXa and thrombin, and

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probably also of FIXa, FXIa, and FXIIa. Subsequently, heparin dissociates from the complex for reuse.[1] This mechanism is responsible for the largest anticoagulant effect. The molecular weight of heparin is variable, and only one-third of UFH molecules possess the unique AT-binding site.

Two additional mechanisms of action have been described. Firstly, at higher concentrations (≥5 units/ml) and in the presence of heparin cofactor II, heparin can catalyse thrombin inhibition without the AT-binding site.[5] Secondly, at very high concentrations, heparin inhibits FXa activation by binding FIXa, independently of AT and heparin cofactor II.[6]

Heparin is administered parenterally as it is not absorbed from the gastrointestinal (GI) tract.[1] Bioavailability with subcutaneous administration is lower compared with intravenous administration (Table 1).[4] Owing to its strong negative charge, UFH molecules associate with the vascular endothelium, releasing proteins, such as lipoprotein lipase that are physiologically attached to endothelial glycosaminoglycans, including heparan sulfate.[7]

Heparin clearance is biphasic: rapid cellular uptake by endothelial cells and macrophages, followed by slower renal elimination. Clearance is dose-dependent, leading to a variable half-life.[8] In addition, the anticoagulant response varies between patients. Therefore, the anticoagulant effect should be monitored using the activated partial thromboplastin time (aPTT) or an anti-Xa assay. In cases where heparin doses are higher (e.g., in cardiopulmonary bypass surgeries), the activated clotting time (ACT) is used. The anticoagulant effect of heparin is rapidly reversed by administration of protamine.[1]

Depolymerising UFH to about one-third of its molecular weight produces LMWH. Smaller LMWH fragments (<18 saccharide units) cannot bind both AT and thrombin simultaneously; thus, LMWH primarily targets FXa.[1]

Compared with UFH, LMWH binds less to plasma proteins and cells, resulting in more predictable pharmacology, a longer half-life, and stable anticoagulant effects (Table 1). Its predictable dose-response allows weight-adjusted dosing in most patients,[6] and reduced endothelial binding enhances subcutaneous bioavailability.[9]

LMWHs are mainly cleared renally; impaired renal function prolongs half-life, sometimes requiring dose adjustment based on peak anti-Xa levels. Their anticoagulant effect is only partially reversed by protamine.[6]

UFH is primarily used in emergent vascular interventions or cardiopulmonary bypass, while LMWH is common for prophylaxis and treatment of venous thromboembolism (VTE), and in

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patients for whom oral anticoagulants are contraindicated, such as during pregnancy or in cancer (Table 2).

Fondaparinux is a synthetic pentasaccharide containing only the AT-binding site, selectively enhancing FXa inhibition. With a half-life of ~17 hours, once-daily subcutaneous dosing achieves therapeutic levels. Its clearance is almost entirely renal, so it is contraindicated in severe renal insufficiency. Monitoring can be done with anti-FXa assays. Although no specific antidote exists, recombinant FVIIa may partially reverse its effect.[4]

Unlike UFH, LMWH, and fondaparinux, direct thrombin inhibitors (DTIs) inhibit thrombin independently of AT. Approved parenteral DTIs include bivalirudin and argatroban. Hirudin, the natural prototype, inspired the shorter synthetic analogue bivalirudin.[4] Bivalent DTIs (hirudin, bivalirudin) bind thrombin at both its active site and exosite 1, while univalent DTIs (argatroban) bind only the active site. DTIs inhibit both soluble and fibrin-bound thrombin.[10]

Bivalirudin is metabolised by proteolysis and hepatic metabolism, though renal impairment may prolong its half-life. Argatroban is primarily metabolised by the liver, requiring dose adjustment in hepatic dysfunction.[4] The anticoagulant effect of DTIs is monitored by aPTT, particularly in patients at high bleeding risk, as no specific antidote is available.[10]

2.2 Vitamin K antagonists

Vitamin K is a coenzyme essential for synthesising vitamin K-dependent coagulation proteins (FII, FVII, FIX, FX, protein C, S, and Z) and several other hepatic and extrahepatic proteins. It undergoes continuous recycling via reduction and oxidation, mediated by the vitamin K epoxide reductase complex subunit 1 (VKORC1). VKAs inhibit VKORC1, inducing functional vitamin K deficiency in hepatocytes and depleting active coagulation factors.[11] Consequently, their anticoagulant effect can be reversed by administering vitamin K. VKAs also affect extrahepatic vitamin K–dependent proteins—17 are known—including osteocalcin, matrix Gla protein (MGP), and growth arrest-specific 6 (Gas6).[12]

VKAs are administered orally. Due to the long half-lives of FII and FX, several days of concurrent heparin are required to reach the narrow therapeutic window.[11] Management of VKAs is challenging due to individual variability and interactions with medications and diet, requiring dose adjustments based on the international normalised ratio (INR), a

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standardised measure of prothrombin time (PT).[13] Fiix-assay studies—suggesting that FII and FX predominantly determine thrombin generation under VKA treatment and that FVII exhibits marked variability—indicate that measuring FII and FX instead of the INR may provide more stable anticoagulation, but this assay is not yet routinely applied.[14, 15]

VKAs are rapidly absorbed in the stomach and small intestine, with almost complete oral bioavailability. Plasma protein binding is high, resulting in a low free plasma concentration that is pharmacologically active.[13, 16] Acenocoumarol has the shortest half-life, followed by warfarin and phenprocoumon (Table 3). The long half-life of phenprocoumon is attributed to slower CYP-mediated clearance, as well as enterohepatic recycling.[16]

Drugs metabolised via CYP2C9 and CYP3A4 may alter VKA plasma concentrations, increasing bleeding risk, or reducing efficacy.[17] As VKAs are primarily cleared hepatically, they are considered safe in end-stage renal disease.[13] In emergencies, anticoagulation can be rapidly reversed with, preferably four-factor, prothrombin complex concentrate (PCC).[17]

2.3 Direct oral anticoagulants

DOACs reversibly bind the active site of FXa (apixaban, edoxaban, rivaroxaban) or FIIa (dabigatran), thereby inhibiting thrombin generation and fibrin formation. Peak plasma concentrations occur within hours of oral intake (Table 3).[18-21]

Dabigatran is a prodrug with low bioavailability due to lipophilicity and pH-dependent absorption; tartaric acid in the formulation enhances uptake.[22] Other DOACs have higher bioavailability due to better solubility and less pH dependence. Half-lives are relatively short, up to 14 hours. Renal excretion varies from 27% for apixaban to 85% for dabigatran.[18-21]

After absorption, DOACs enter the mesenteric circulation or are pumped back into the intestinal lumen by P-glycoprotein (P-gp) transporters.[24] These efflux pumps, expressed in tissues with excretory or barrier functions, limit absorption, enhance elimination, and prevent distribution to sensitive organs like the brain.[25] Cytochrome P450 enzymes also play an important role in the clearance of apixaban and rivaroxaban.[26] Consequently, co-administration with strong P-gp or CYP3A4 inducers or inhibitors can lead to subclinical concentrations or increased bioavailability and subsequent thrombosis or bleeding risk, respectively, and is therefore contraindicated.[26] Apixaban and rivaroxaban are also substrates of the breast cancer resistance protein, which functions similarly to P-gp.[27, 28]

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Routine laboratory monitoring and dose adjustment are not necessary given their stable pharmacokinetics. However, DOAC concentrations can be measured directly via mass spectrometry or indirectly using anti-FXa assays (for FXa inhibitors) or diluted thrombin time (for dabigatran).[29, 30] Several point-of-care assays are currently under investigation for emergency use.[31]

Despite fixed dosing, DOAC plasma concentrations can vary substantially between individuals, potentially increasing thrombosis or bleeding risk when plasma levels fall outside the "on-therapy" ranges as established in major clinical trials.[32] Monitoring may be useful in selected patients—frail elderly, those with impaired GI absorption, extreme weight, or potential drug interactions—where concentrations may deviate from expected ranges.[17, 33]

Reversal agents include andexanet alfa (for apixaban, rivaroxaban, off-label edoxaban) and idarucizumab (for dabigatran). Andexanet alfa is a recombinant, inactive FXa; idarucizumab is a monoclonal antibody binding dabigatran with high affinity.[23] PCCs are often used instead due to cost and safety concerns, as andexanet alfa has been linked to increased thrombotic events in bleeding patients with underlying thromboembolic risk.[34, 35] Only dabigatran can be removed by haemodialysis due to its low plasma protein binding (35%).[26]

Oral anticoagulants are mostly prescribed for the treatment and prevention of thrombosis. Large clinical trials have established DOACs' comparable or superior efficacy and lower bleeding risk compared to the previous gold standard, VKAs, for the major indications atrial fibrillation (AF) and VTE.[36] However, VKAs remain preferred in high thrombogenic conditions like mechanical heart valves, AF associated with mitral valve disease, antiphospholipid syndrome (APS, especially with triple-positive antibodies), and recurrent thrombosis while on DOACs (Table 2). In frail elderly, there is uncertainty regarding the optimal anticoagulant for stroke prevention in AF; for patients who are stable on VKAs, it may be prudent not to switch to a DOAC.[37, 38]

2.4 Factor XI-inhibitors

Inhibiting FXI(a) can reduce thrombus formation via the contact activation pathway without inhibiting physiological haemostasis mediated via the tissue factor (TF) pathway. Both oral and parenteral compounds are undergoing clinical testing.[39]

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FXI inhibitors fall into three main classes: small molecules, antisense oligonucleotides, and monoclonal antibodies. Others include natural peptides and aptamers. Small molecules are orally administered reversibly binding FXIa. Due to their short half-life, daily dosing is required to maintain a therapeutic effect.[3]

Antisense oligonucleotides are given subcutaneously and act in the liver by degrading FXI mRNA, thereby inhibiting protein synthesis.[40] The anticoagulant effect develops over several weeks, but their long half-life allows weekly or monthly dosing, improving adherence yet complicating rapid reversal. In emergencies, replacement with fresh frozen plasma or FXI concentrate may be considered.[3] Their strong plasma protein binding limits renal clearance.[40]

Monoclonal antibodies, administered intravenously or subcutaneously, selectively bind FXI(a) with rapid onset and prolonged half-life, enabling extended dosing intervals. They are metabolised and cleared mainly by phagocytic cells and the reticuloendothelial system. For urgent reversal, recombinant FVIIa may be used; however, available data suggest an increased risk of thromboembolic events in off-label settings—particularly arterial thromboses—though most trials have been underpowered to reliably quantify this risk.[41]

3. Intended effects

3.1 Anticoagulant and antithrombotic mode of action

All anticoagulants are administered to prevent fibrin formation and, consequently, limit thrombosis, allowing natural fibrinolytic mechanisms to restore reperfusion of the occluded blood vessel(s) (Figure 1). Different anticoagulants achieve inhibition of fibrin formation by different routes, with the common denominator being the inhibition of thrombin formation. Traditionally, clotting assays have been used to study the impact of anticoagulants in plasma, with the aPTT and PT serving as the most sensitive tests for UFH and VKAs, respectively. LMWHs, DTIs, and DOACs have variable effects on clotting assays.

Thrombin generation analysis in plasma has been helpful in clarifying mechanistic differences among anticoagulants. All FXa inhibitors and dabigatran prolong the lag time and time to peak in thrombin generation assays, but only the FXa inhibitors lower the thrombin peak level in a nonlinear, dose-dependent manner.[42] Within the spectrum of anticoagulants, VKAs appear to have the strongest potential to inhibit thrombin generation in a dose-dependent way, probably because VKAs affect both the extrinsic and intrinsic routes of

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coagulation. Clinical studies confirm this overall stronger suppression of thrombin generation, based on lower levels of the prothrombin fragment F1+2 in patients on VKAs versus DOACs.[43, 44]

Additionally, UFH and, to some extent, LMWH are generally more potent in inhibiting thrombin generation than selective anticoagulants, including DOACs.[45] This may be explained by the fact that UFH, like VKAs, inhibits the coagulation cascade at multiple levels. These differences in modes of action may also underlie the superior efficacy of VKAs and heparins in highly thrombogenic settings like APS, mechanical heart valves, and extracorporeal circuits.

3.2 Effects on fibrin clot formation and fibrinolysis

Data on the effects of heparins on fibrin clot properties and fibrinolysis remain relatively limited. In patients with acute pulmonary embolism treated with enoxaparin, higher plasma anti-FXa activity was associated with increased clot permeability and shortened clot lysis time.[46] Another study demonstrated that heparin and LMWH, but not fondaparinux, modified fibrin clot structure, potentially enhancing fibrinolysis.[47] In contrast to other anticoagulants, fondaparinux increased clot permeability, without dose dependent effect.[48] The absent or modest effect of fondaparinux may be attributed to its selective inhibition of FXa. This hypothesis was tested in an in vitro study, which demonstrated that profibrinolytic activity was more closely associated with thrombin inhibition than with anti-FXa activity.[49] Thrombin exerts antifibrinolytic effects through activation of thrombin activatable fibrinolysis inhibitor (TAFI); inhibition of thrombin activity renders fibrin clot structure more sensitive to fibrinolysis.[49]

Given that fibrin clot formation is dependent on thrombin availability, inhibition of thrombin generation may result in alterations of fibrin clot properties.[50] Fibrin clots generated in the presence of rivaroxaban and dabigatran were characterised by thicker fibrin fibers and increased permeability compared to clots generated in their absence. In addition, reduced activation of TAFI also contributed to the formation of more lysable clots.[51, 52] Furthermore, rivaroxaban and apixaban have been shown in vitro to shorten clot lysis time, possibly by enhancing t-PA cofactor activity and promoting plasmin generation.[53]

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4. Unintended effects

4.1 Impairment of haemostasis and bleeding risk

Clinically, bleeding remains the major and most feared adverse effect of anticoagulant therapy. To date, no anticoagulant completely lacks bleeding-enhancing potential, although inhibition of FXI(a) is anticipated to offer a safer profile. The mechanisms underlying anticoagulant-associated bleeding are still incompletely understood. Established patient-related risk factors include advanced age, renal impairment, concomitant medications, a history of bleeding or stroke, and hypertension; these have been incorporated into various bleeding risk scores.[54]

The quality of anticoagulant management also plays a critical role. For VKAs, the time in therapeutic range (TTR) remains a key determinant of both efficacy and bleeding risk.[54] In DOAC users, variability in peak plasma levels is associated with bleeding, while trough levels correlate with thrombotic risk.[17, 32] For all anticoagulants, there is a dose-dependent increase in bleeding risk, highlighting the need to identify the optimal intensity for each patient and indication.

Certain types of bleeding are more often associated with specific classes of anticoagulants. For example, DOACs are associated with a smaller risk of intracranial haemorrhage compared to VKAs, which is possibly related to the multiple target effect of VKAs including on FVII inhibition in the TF-rich brain.[55, 56] Conversely, GI bleeding occurs relatively more often with DOACs (dabigatran, edoxaban, and rivaroxaban) than with VKAs or LMWHs, likely due to their variable concentration of active drug in the GI tract and higher local tissue penetration.[55] Two out of three women initiating anticoagulation for VTE experience abnormal menstrual bleeding.[57] Apixaban, dabigatran, and VKAs appear to carry a lower risk of heavy menstrual bleeding compared to rivaroxaban and edoxaban.[58] However, prospective randomised studies directly comparing different anticoagulants are still lacking.

One underexplored bleeding mechanism relates to endothelial barrier integrity, which may be critically important in situations where this barrier is perturbed, such as in ischemia-reperfusion injury following ischemic stroke.[59] Similarly, in GI or other organ bleeding, local endothelial damage may be a very important contributor or cause of bleeding during anticoagulation. As endothelium integrity is affected by inflammation, oxidation, and trauma, related biomarkers (e.g., GDF-15, MMP-9, vascular adhesion protein-1, serum

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protein S100B, fibronectin, and tight junction proteins like occludin) may help to better estimate bleeding risk.[59] Furthermore, biomarkers related to inflammation, vascular remodelling, endothelial cell damage, coagulation, and fibrinolysis (specifically GDF-15, cTnT-hs, and seven novel biomarkers) have been independently associated with major bleeding in patients with AF receiving OAC.[60]

4.2 Interactions with blood platelets

Given the inherent risk of bleeding associated with anticoagulation, numerous studies have explored potential interactions with platelet count and function. Thrombocytopenia, as frequently observed in patients receiving cancer treatment, increases bleeding risk and necessitates adjustment of anticoagulant management.[61] Although anticoagulant associated thrombocytopenia is uncommon, the rare occurrence of heparin-induced thrombocytopenia (HIT) remains a serious and feared complication.[62] The risk of HIT is lower with LMWH and absent with fondaparinux compared with UFH, reflecting reduced binding of shorter heparin chains to platelet factor 4.[6] Given the diagnostic and therapeutic complexity of HIT, detailed discussion is provided in dedicated expert reviews.[63, 64]

Apart from HIT, UFH may, possibly through electrostatic interactions with surface proteins, enhance platelet aggregation, depending on heparin chain length and its affinity for AT.[65, 66] However, these effects are generally of limited clinical relevance, except in the context of extracorporeal circulation, where UFH exposure may contribute to platelet count reduction and functional impairment as part of the broader coagulopathy associated with such procedures.[67]

All anticoagulants affect thrombin-induced platelet activation, one of the most relevant haemostatic pathways in vivo. Thrombin is a principal agonist of platelet activation via binding to and activation of PAR-1 and PAR-4, with different kinetics.[68, 69]

Several studies have examined platelet activation pathways in patients receiving warfarin. Platelet aggregation in response to collagen or epinephrine appears unaffected by warfarin therapy and may even be enhanced at higher INR levels.[71, 72] Increased platelet reactivity has also been reported in patients treated with warfarin for AF or secondary stroke prevention.[73] In healthy volunteers exposed to subtherapeutic warfarin levels (INR ≈1.5) for 35 days, heightened platelet responses to adrenaline and adenosine diphosphate (ADP), and to a lesser extent to collagen, were observed.[74] Whether this enhanced platelet responsiveness

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represents a compensatory mechanism in the context of suppressed thrombin-mediated platelet activation remains uncertain.

In platelet-rich plasma, both FXa inhibitors and DTIs inhibit TF-triggered platelet aggregation in a concentration-dependent manner.[70] The effect of rivaroxaban on platelet function has been the focus of several studies.[75] In vitro, rivaroxaban inhibited P-selectin expression at the platelet surface upon stimulation of aggregation by thrombin (or TF).[76] Additionally, FXa inhibition reduced soluble glycoprotein VI (GPVI) concentrations, thereby limiting platelet activation during atherosclerotic plaque rupture, and diminished thromboxane synthesis and Nox-2-dependent oxidative stress following GPVI stimulation.[77]

In vivo, rivaroxaban (and apixaban) did not alter platelet responses to a range of agonists in patients with AF.[78, 79] Several clinical studies have shown reduced levels of soluble platelet activation biomarkers, including P-selectin and β-thromboglobulin, in patients treated with rivaroxaban;[80-82] however, other studies did not confirm this effect.[79] Rivaroxaban reduced platelet activation, aggregation, and thrombus formation through inhibition of FXa-mediated PAR-1 activation under flow conditions and in vivo. This resulted in the formation of less stable thrombi and was associated with increased thrombus fragment embolisation in murine models of arterial thrombosis.[83] Finally, there is also evidence suggesting that rivaroxaban inhibits thromboxane production in vivo.[84]

Long-term FXa-inhibition by DOACs has been associated with reduced platelet-mediated thromboinflammation and, consequently, smaller infarct sizes after acute myocardial infarction and stroke. Long-term treatment induces alterations in platelet RNA and proteomic profiles that attenuate platelet granule secretion and thereby reduce neutrophil extracellular trap (NET) formation. This effect was verified by single-gene analysis demonstrating downregulation of vesicle-associated membrane protein 8 (VAMP-8).[85]

The focus on platelet reactivity increased after the RE-LY study observed higher myocardial infarction rates in patients with AF taking dabigatran compared to warfarin.[86, 87] Several studies addressed effects of dabigatran on platelets either in blood from healthy subjects with neutral effects, or in patients with stroke on dabigatran who had higher platelet reactivity to ADP as compared to patients on FXa inhibitors.[88, 89] A study in hospitalised patients showed increased platelet reactivity after starting dabigatran, associated with increased PAR density at the platelet surface.[90] In vitro, addition of a DTI to whole blood reduced the amount of thrombin-induced platelet binding to monocytes or granulocytes, expressing a reduced amount of TF mRNA.[91] Although dabigatran did not influence PAR-1 expression on platelets from AF patients, expression of the platelet activation marker CD62P was inhibited after thrombin stimulation.[92] At the same time, chronic dabigatran therapy has been associated with increased thrombin receptor-activating peptide (TRAP)-induced

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platelet aggregation, as well as enhanced platelet adhesion and thrombus formation due to altered GPIbα interactions.[77] Overall, these, sometimes conflicting, data do not suggest any consistent effects of dabigatran on platelet reactivity, suggesting that other factors may explain the unexpected higher risk of myocardial infarction. One speculated mechanism is that DTIs impair thrombin-induced protein C activation, causing a temporary hypercoagulability, similar to what may rarely occur with VKAs, but the clinical relevance in this context is unknown.[93]

Given the beneficial effects on clinical outcomes of dual pathway inhibition (DPI) with low-dose aspirin and low-dose rivaroxaban (2.5 mg bd) in the COMPASS and VOYAGER trials,[94, 95] mechanistic studies addressed the addition of low-dose rivaroxaban to antiplatelet therapy. These studies reported reduced thrombus formation without effects on pure platelet-dependent clotting or platelet aggregation measured by light transmission aggregometry. However, the addition of rivaroxaban reduced platelet expression of high mobility group box 1 (HMGB-1), without a concurrent effect on P-selectin.[96]

In conclusion, most, if not all, OACs appear to inhibit thrombin-induced platelet activation and aggregation, whereas platelet activation by other agonists such as ADP or collagen remains unchanged or may even be enhanced. This phenomenon could represent a compensatory response to reduced thrombin-mediated signalling. Subtle additional platelet-inhibitory effects of certain DOACs, such as rivaroxaban, may contribute not only to their antithrombotic efficacy but also to the increased bleeding risk observed with DPI.

4.3 Impact on extracellular vesicles and immune modulation

Extracellular vesicles (EVs) are circulating membrane-derived particles with a messenger function, released by cells in response to activation or apoptosis. In patients with stable cardiovascular disease, DPI was associated with a reduction in EVs from activated platelets, less so from neutrophils, and not from endothelial origin.[97] Characterisation of EV contents showed a subset of differentially expressed proteins, including several of platelet and neutrophil origin, and an overall reduction in inflammasome activation compared to aspirin alone.[97]

In patients with AF, treatment with rivaroxaban compared to warfarin was associated with a less inflammatory proteomic profile in EV-enriched plasma fractions, along with less endothelial activation markers.[98] Similarly, EVs from patients with VTE managed with either warfarin or rivaroxaban showed a differential expression of six proteins (PROZ, F2,

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SERPINA10, APCS, F10, and PROS1), with an anticoagulant and anti-inflammatory profile in the DOAC treated subjects.[99]

Potentially, a reduction in EV inflammatory proteome contents, may reflect or have impact on systemic inflammatory activity. In a prospective observational study of patients with stable atherosclerotic disease, DPI was associated with reductions in plasma levels of IL-6 and fibrinogen after 24 weeks, compared to baseline.[100] However, in patients with diabetic atherosclerotic disease, addition of low dose rivaroxaban did not have any detectable effect on endothelial and inflammatory biomarkers.[101] Moreover, three months DPI treatment in other patients with stable atherosclerotic disease did not reveal any relevant effects on circulating inflammatory mediators or mononuclear cell responsiveness.[102]

In a study comparing AF patients using therapeutic dosed rivaroxaban with warfarin, a trend towards a reduced level of pro-inflammatory cytokines and an increase in chemokines in rivaroxaban treated subjects was seen, with uncertain net anti-inflammatory effect.[103] In a substudy of the X-VeRT trial on cardioversion in AF patients, biomarkers of coagulation and inflammation were similarly lowered under treatment with both rivaroxaban and VKAs, though F1+2 levels remained more elevated under rivaroxaban.[44] Finally, using a panel of inflammatory markers, 187 patients with AF randomly assigned to rivaroxaban or dabigatran did not yield any differences between these drugs at 12 months follow up.[104] Summarising, subtle indications of anti-inflammatory effects of DOACs (mostly rivaroxaban) were found in circulating EVs without evident changes in systemic inflammatory biomarkers; a similar effect, if any, of VKAs or dabigatran cannot be excluded.

4.4 Endothelium

Vascular endothelial cells play a central role in regulating coagulation by maintaining a balance between procoagulant and anticoagulant mechanisms.[105] This regulation occurs in part through the uptake and metabolism of endogenous glycosaminoglycans, as well as the uptake of exogenous compounds such as heparin.[106] The structural characteristics of heparin provide multiple potential binding sites for proteins containing heparin-binding domains, facilitating primarily electrostatic interactions that extend beyond coagulation pathways.[107]

UFH and, to some extent, LMWH associate with positively charged proteins at the negatively charged glycocalyx, competitively displacing molecules such as lipoprotein lipase, TF pathway inhibitor, and platelet factor 4.[108] In addition, heparins display diverse anti-inflammatory effects by binding to interleukin-6 (IL-6), the IL-6/IL-6 receptor complex,

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interferon-gamma, and other inflammatory proteins, including chemokines, cytokines, and complement factors.[109] The overall effect of heparin administration is thought to include anti-inflammatory actions; however, these properties have not been shown in clinical trials to confer additional therapeutic benefit in the management of thromboinflammatory diseases.

An essential endogenous anticoagulant pathway is the protein C system, which is predominantly activated at the endothelial surface through the binding of thrombin to thrombomodulin and endothelial cell protein C receptor (EPCR) (Figure 2). Activated protein C (APC) exerts anti-inflammatory, cytoprotective, and endothelial barrier–stabilizing effects, thereby preventing endothelial cell injury and death.[110, 111] As noted earlier, initiating VKAs can induce a transient hypercoagulable state due to a relative depletion of APC, particularly in individuals with a congenital or acquired protein C deficiency. Whether the generation of APC, and consequently its endothelium protective actions, is differentially regulated by inhibition of thrombin generation via FXa inhibitors or DTIs, as compared to VKAs, is unknown. It is conceivable that residual thrombin activity may continue to support protein C activation, or that alternative APC/EPCR/PAR signalling mechanisms contribute to maintaining endothelial integrity.[112]

VKA-induced uncarboxylated proteases FXa and thrombin, still retain catalytic activity against PARs, although with less efficiency given the impaired phospholipid localisation. Consistent with this, recombinant Gla-domainless FXa and thrombin have been shown to activate PARs, and similarly, both APC and Gla-domainless APC can activate PAR-1 in human umbilical vein endothelial cells (HUVECs), provided that the receptor is not shielded by lipid raft localization. Whether such protease–receptor interactions persist and remain physiologically relevant in vivo during VKA therapy remains to be determined.[113]

The effects of DOACs on endothelial cells have attracted significant research interest.[114] In in vitro studies with HUVECs, oxysterol reduced endothelial cell integrity and promoted inflammation. Treatment with rivaroxaban or dabigatran, however, appeared to restore endothelial barrier function by upregulating endothelial adhesion molecules such as vascular endothelial (VE)-cadherin, while simultaneously attenuating inflammation through downregulation of intercellular adhesion molecule-1 (ICAM-1) expression.[115, 116] Similarly, in an in vitro model of uraemia, apixaban prevented endothelial dysfunction and exhibited anti-inflammatory and antioxidant properties.[117] An in vitro study of human brain endothelial cells showed that thrombin-mediated cleavage of PAR-1 altered the expression of tight junction proteins towards a negative effect on endothelial cell integrity. PAR-1 cleavage by thrombin

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was diminished by pretreatment with dabigatran and rivaroxaban, but not by warfarin or heparin.[118]

As discussed previously, FXa and thrombin interact with PAR-1 and PAR-2 at the vascular endothelium, with their effects shaped by local enzyme concentration and the functional reserve of natural anticoagulant receptors thrombomodulin and EPCR (see previous discussion on mechanisms in [119]). Recent data show that rivaroxaban attenuates NLRP3 inflammasome activation via PARs, MAPK, and NF-κB pathways in a diabetic mouse model and in isolated arterial endothelial cells.[120] Additional anti-atherogenic effects may include reversal of FXa-mediated suppression of macrophage autophagy, as demonstrated in an atherosclerosis mouse model.[121]

Several in vivo studies have explored the effects of DOACs on endothelial function. One RCT involving patients with type 2 diabetes and high cardiovascular risk reported that low dose rivaroxaban improved endothelial function, as assessed by enhanced post-ischaemic forearm blood flow, a numerical increase in skin blood flow, and reduced soluble P-selectin compared with aspirin alone. Both groups showed a significant increase in platelet-derived microparticles (PMPs), but PMPs from rivaroxaban-treated patients stimulated HUVEC proliferation more than aspirin. However, no differential effects on biomarkers of inflammation, endothelial cell activation, or vascular stiffness measurements, were noted.[122] Another study in individuals with chronic heart failure and AF observed improved endothelial function after switching from VKAs to DOACs, as assessed by flow-mediated dilation.[123] Together, these in vitro and in vivo findings indicate that DOACs may contribute to endothelial cell integrity by reducing endothelial and inflammatory activation.

FXI may exert distinct effects on the endothelium. Endothelial barrier integrity is regulated by the endothelial cell-specific proteins ROBO-4 and VE-cadherin. FXIa enhances cleavage of VE-cadherin by the metalloproteinase ADAM10, thereby enhancing loss in endothelial cell barrier function. Theoretically, inhibiting FXIa may be beneficial in supporting VE-cadherin function under stress conditions like sepsis.[124]

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5. Impact of anticoagulants in the vessel wall and the heart

5.1 Venous thrombosis and post-thrombotic syndrome

Following inflammatory endothelial damage associated with deep vein thrombosis (DVT), re-endothelialisation occurs as part of the thrombus resolution process. In a rat model of DVT, treatment with LMWH was shown to significantly accelerate re-endothelialisation during the early phase of thrombus resolution.[125] Supporting these findings, another study investigating DVT in rats reported that LMWH administration resulted in a twofold reduction in vein wall stiffness compared to untreated controls.[126] In earlier work, both LMWH administration and P-selectin inhibition were associated with decreased vein wall stiffness.[127] However, it remains unclear whether these effects are mechanistically related. Notably, LMWHs have been shown to stabilise the endothelial glycocalyx and inhibit its shedding during inflammation.[128] By preserving endothelial integrity, LMWHs reduce leukocyte adhesion and vascular permeability, thereby contributing to vascular protection and improved thrombus resolution.

As noted earlier, DOACs may beneficially modify both the structural and functional characteristics of fibrin clots, thereby promoting thrombus resolution. These anticoagulant effects may contribute to the prevention of post-thrombotic syndrome (PTS), a complication of DVT in which residual vein obstruction (RVO) is thought to play a pathophysiological role.[53] PTS develops as a consequence of persistent venous obstruction, venous hypertension, inflammation, and endothelial activation.[129] In patients with PTS, fibrin clot permeability is reduced and clot lysis time is prolonged compared with patients with DVT who do not develop PTS.[130] These findings suggest that the modulation of fibrin clot properties and the shortening of clot lysis time by DOACs may confer clinical benefits by reducing the risk of PTS in patients with DVT.

Interestingly, despite similar rates of RVO, the incidence of PTS was lower in patients treated with DOACs compared to those receiving VKAs in two cohort studies.[131, 132] This observation indicates that the efficacy of DOACs in preventing PTS may involve additional mechanisms beyond the reduction of RVO. A plausible explanation lies in the pathophysiology of PTS itself, which is driven by persistent inflammation leading to endothelial activation and dysfunction.[129] Given the anti-inflammatory properties of DOACs, these agents may indirectly preserve endothelial function and thereby contribute to their beneficial impact on PTS prevention.

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5.2 Atherosclerosis and atherothrombosis

Although the anti-inflammatory properties of LMWHs make them an attractive therapeutic option in arterial vascular disease, their requirement for parenteral administration limits long-term use, except in specific circumstances such as pregnancy.

Long-term use of VKAs and DOACs reduces the vascular burden associated with excessive thrombin generation. However, VKAs may promote arterial medial calcification by inhibiting the γ-carboxylation of vascular smooth muscle cell–derived proteins. In contrast, more potent suppression of PAR-1 activation through thrombin inhibition in endothelial cells has been proposed as a mechanism by which DOACs may prevent vascular calcification. By selectively inhibiting FXa and thrombin, DOACs attenuate PAR-1–mediated vascular smooth muscle cell migration and proliferation.[133]

Several observational studies have linked VKA use with an increased risk of vascular calcification and greater coronary plaque burden.[134-136] This is driven by VKA-induced reductions in vitamin K, which serves as a cofactor for the transformation of vitamin K-dependent proteins. For MGP in particular, impaired γ-carboxylation diminishes its ability to inhibit calcium deposition and crystallisation within the vessel wall.[12]

One study reported a higher incidence of vascular calcification in patients on warfarin or rivaroxaban compared to those without anticoagulation, while no such increase was seen on dabigatran.[137] Moreover, two randomised trials associated VKA use with coronary plaque progression over 52 weeks, compared to apixaban and rivaroxaban.[138, 139] In contrast, another randomised trial found no significant difference in coronary calcification progression between VKAs and rivaroxaban after 12 and 24 months.[140] Collectively, these findings suggest that DOACs may confer an advantage over VKAs in mitigating vascular calcification. However, it remains uncertain whether this effect is consistent across different vascular territories or whether specific DOACs, such as dabigatran, offer superior protection compared with FXa inhibitors.

As discussed in previous articles, also in this journal, DOACs exert notable effects on the vasculature, particularly by inhibiting the progression or promoting the regression of atherosclerosis in experimental mouse models.[119, 141, 142] Furthermore, interference with the protease-PAR signalling properties of FXa and thrombin may profoundly influence smooth muscle cell function, thereby modulating the balance between physiological vascular repair and pathological remodeling.[143] Owing to their small molecular size and substantial volume

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of distribution, particularly for dabigatran, DOACs may penetrate tissues efficiently and exert significant extravascular effects.

The concept of tissue-penetrating properties is exemplified in tumour models, where tumour-associated macrophages have been shown to produce FXa, thereby activating local FXa–PAR-2 signalling pathways that promote tumour growth and progression.[144] Inhibition of this pathway may, in turn, reprogram tumour-associated macrophages, enhancing tumour antigen presentation and facilitating immune-mediated tumour cell destruction in certain experimental models.[145, 146]

These examples illustrate the broader principle that tissue function and repair can be modulated by protease inhibitors such as DOACs. Additional studies have demonstrated intracellular effects beyond anticoagulation, including dabigatran in experimental models of asthma and Alzheimer's disease, rivaroxaban influencing mitochondrial proteins in models of aortic aneurysm, and dose-dependent antioxidant effects of both rivaroxaban and edoxaban in mitochondria derived from renal cells.[147-151] Recent prospective data aligns with these mechanistic insights, showing that patients with AF and chronic kidney disease experienced fewer adverse kidney outcomes when treated with rivaroxaban compared with VKAs.[152] It remains uncertain whether this difference reflects VKA-associated vascular injury and calcification, anti-inflammatory effects of rivaroxaban, or an interplay of both mechanisms.

Clinically, evidence for vascular pleiotropic effects of DOACs are still limited. A retrospective analysis suggested reduced restenosis rates following femoral-popliteal artery stenting in patients receiving DPI compared to those on DAPT. Another small cohort study, using low dose rivaroxaban with clopidogrel instead of aspirin, also reported improved patency and reduced restenosis, without impact on clinical outcomes like amputation rates.[153, 154]

Therapeutic doses of rivaroxaban or dabigatran did not significantly affect intima media thickness or biomarkers of atherosclerosis and endothelial dysfunction compared to warfarin after 12 or 24 months in patients with AF.[155] Switching from prolonged VKA therapy to rivaroxaban led to a reduction in brachial-ankle pulse wave velocity after three months, compared to continued VKA use.[156] However, flow mediated dilation of the brachial artery in patients on long-term VKA or apixaban therapy showed no significant differences between groups.[157]

Taken together, these limited data from small human studies with relatively short follow-up periods do not provide clear evidence for vascular protective effects of DOACs at either the macrovascular or microvascular level. Nonetheless, such effects may emerge after prolonged exposure or may be restricted to individuals with specific characteristics, such as

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enhanced inflammation or underlying dyslipidaemia, comparable to atherogenic mouse models used in preclinical DOAC research.

Similarly to DOACs, targeting FXI with either a monoclonal antibody (14E11) or an antisense oligonucleotide reduced atherosclerosis burden in mouse studies. In addition, 14E11 prevented disruption of endothelial cell VE-cadherin in the aortic sinus, suggesting a FXI-specific effect, although a contribution of reduced thrombin generation cannot be excluded.[158] One specific feature may be that, in mice, most FXI is associated with the endothelial glycocalyx, implying a greater local functional effect than in larger species like baboons and humans.[159] Nevertheless, given that FXI may be involved in diverse inflammatory mechanisms at or beneath the endothelial cell surface in humans, interfering in the function of this protein may have wider implications beyond inhibition of the downstream intrinsic cascade.[160]

5.3 The heart

VKAs remain the preferred OAC for high risk thrombosis patients, which may be due to a more potent anticoagulant effect.[161] Another specific VKA effect involves the reduced synthesis of vitamin K-dependent proteins in tissues, including Gas6.[162] VKAs inhibit the gamma-carboxylation of Gas6, a ligand for the TAM family of tyrosine kinase receptors.[163, 164]

The Gas6/Axl pathway may be implicated in remodelling processes in cardiac and vascular tissues.[164] In Gas6 knockout mice, chronic pressure overload induced by aortic banding resulted in reduced cardiac hypertrophy, fibrosis, and contractile dysfunction compared to wild-type controls, whereas Gas6 overexpression induced an adverse phenotype. In hypertrophic murine hearts, Gas6 expression was upregulated in comparison with control hearts. Similarly, Gas6 expression was upregulated in the hearts of patients with dilated cardiomyopathy compared to donor hearts.[165] The DRAGON-HF trial linked elevated plasma Gas6 concentrations in patients with acute heart failure to a higher all-cause and cardiovascular mortality.[166]

Taken together, a better understanding of VKA's pleiotropic effects is important to inform future therapeutic strategies targeting specific intermediates in the Gas6 pathway.

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In an in vitro study with human cardiac fibroblasts, incubation with FXa led to overexpression of pro-inflammatory genes via PAR-1, suggesting DOACs may exert cardioprotective effects by inhibiting FXa/thrombin dependent PAR-1 activation.[167] Ischemia-reperfusion injury after myocardial infarction is characterised by an inflammation response and apoptosis, involving a complex and critical interaction between inflammation and coagulation. In animal models, natural anticoagulants have demonstrated cardioprotective functions.[168] Compared to untreated controls, mice treated with rivaroxaban and dabigatran showed reduced infarct sizes after induction of ischemia-reperfusion injury, but only rivaroxaban showed an anti-inflammatory gene expression profile.[169]

Experiments have linked FXI to cardiac function, which is relevant given the ongoing developments of FXI inhibitors. In a mouse model of diastolic heart failure, FXI overexpression in the liver improved diastolic function, whereas FXI knockout mice exhibited increased severity. This effect of FXI may be related to bone morphogenetic protein 7 (BMP7), which is present in significant concentrations within cardiac tissue. FXIa cleaves BMP7 in the extracellular matrix in the heart, subsequently activating the BMP7-SMAD1/5 signalling pathway. As a result, genes involved in inflammation and fibrosis are inhibited.[170] In a large community-based cohort study, lower plasma FXI levels were associated with a higher incidence of heart failure, particularly among participants aged ≥75 years.[171] This association was validated in the Cardiovascular Health Study.[171] However, it remains unclear whether this age-related risk is confounded by other risk factors for heart failure or if the effect of decreased plasma FXI levels depends on duration of exposure.

One risk factor both for atherosclerosis and heart failure is hypertension. Translational studies showed that platelet-localised FXI plays a role in accelerated thrombin generation, associated with and possibly causally involved in uncontrolled hypertension.[172] Although a link between plasma FXI(a) levels and hypertension has not yet been reported from clinical trials of FXI inhibitors, recent research is exploring FXI inhibition as a possible antihypertensive strategy.[173]

6. Concluding remarks

Over the years, anticoagulant therapy has evolved significantly, primarily focusing on preventing and treating thrombosis by inhibiting thrombin generation and subsequent fibrin formation. Beyond their established anticoagulant effects and associated bleeding risks, accumulating evidence highlights important pleiotropic effects both outside and inside the

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cardiovascular system, involving interactions with platelets, leukocytes, cytokines, chemokines, EVs, endothelial cells, VSMCs, endothelial receptors, and signalling pathways.

Overall, the dynamic landscape of anticoagulation therapy reveals that, in addition to their intended anticoagulant targets, pleiotropic effects contribute to therapeutic outcomes. Long-term data on pleiotropic effects of DOACs and FXI inhibitors are still limited, highlighting the need for further research to determine their clinical implications and whether certain patient populations may benefit from specific pleiotropic effects of anticoagulants.

Conflict of Interest: HtC is consultant for Synapse Research Institute, a diagnostic R & D company and shareholder with Coagulation Profile, a Maastricht University spinoff small diagnostic company.

References

1. Hirsh J, Bauer KA, Donati MB, Gould M, Samama MM, Weitz JI. Parenteral anticoagulants: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines (8th Edition). *Chest* 2008;**133**:141s-159s.
2. Cranenburg EC, Schurgers LJ, Vermeer C. Vitamin K: the coagulation vitamin that became omnipotent. *Thromb Haemost* 2007;**98**:120-125.
3. Occhipinti G, Laudani C, Spagnolo M, Finocchiaro S, Mazzone PM, Faro DC, Mauro MS, Rochira C, Agnello F, Giacoppo D, Ammirabile N, Landolina D, Imbesi A, Sangiorgio G, Greco A, Capodanno D. Pharmacological and clinical appraisal of factor XI inhibitor drugs. *Eur Heart J Cardiovasc Pharmacother* 2024;**10**:245-258.
4. Olie RH, ten Cate H, De Caterina R, Moliterno D, Kristensen S. 87Heparins, fondaparinux, hirudin, bivalirudin, argatroban, and danaparoid. In: De Caterina R, Moliterno DJ, Kristensen SD, eds. The ESC Textbook of Thrombosis: Oxford University Press, 2023:0.
5. Tollefsen DM, Majerus DW, Blank MK. Heparin cofactor II. Purification and properties of a heparin-dependent inhibitor of thrombin in human plasma. *J Biol Chem* 1982;**257**:2162-2169.
6. Garcia DA, Baglin TP, Weitz JI, Samama MM. Parenteral anticoagulants: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest* 2012;**141**:e24S-e43S.
7. Shimada K, Gill PJ, Silbert JE, Douglas WH, Fanburg BL. Involvement of cell surface heparin sulfate in the binding of lipoprotein lipase to cultured bovine endothelial cells. *J Clin Invest* 1981;**68**:995-1002.

ACCEPTED MANUSCRIPT

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8. Boneu B, Caranobe C, Sie P. Pharmacokinetics of heparin and low molecular weight heparin. *Baillieres Clin Haematol* 1990;**3**:531-544.
9. Samama MM, Gerotziafas GT. Comparative pharmacokinetics of LMWHs. *Semin Thromb Hemost* 2000;**26 Suppl 1**:31-38.
10. Di Nisio M, Middeldorp S, Büller HR. Direct thrombin inhibitors. *N Engl J Med* 2005;**353**:1028-1040.
11. Ansell J, Hirsh J, Hylek E, Jacobson A, Crowther M, Palareti G. Pharmacology and management of the vitamin K antagonists: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines (8th Edition). *Chest* 2008;**133**:160s-198s.
12. Wen L, Chen J, Duan L, Li S. Vitamin K-dependent proteins involved in bone and cardiovascular health (Review). *Mol Med Rep* 2018;**18**:3-15.
13. Stehle S, Kirchheiner J, Lazar A, Fuhr U. Pharmacogenetics of oral anticoagulants: a basis for dose individualization. *Clin Pharmacokinet* 2008;**47**:565-594.
14. Ingason AB, Gudmundsdottir BR, Palsson R, Hreinsson JP, Lund SH, Letertre LR, Rumba E, Agustsson AS, Bjornsson ES, Onundarson PT. Low incidence of thromboembolism with Fiix-monitored warfarin compared to conventional warfarin and DOACs in patients with AF. *Blood Vessel Thromb Hemost* 2025;**2**:100056.
15. Jonsson PI, Letertre L, Juliusson SJ, Gudmundsdottir BR, Francis CW, Onundarson PT. During warfarin induction, the Fiix-prothrombin time reflects the anticoagulation level better than the standard prothrombin time. *J Thromb Haemost* 2017;**15**:131-139.
16. Ufer M. Comparative pharmacokinetics of vitamin K antagonists: warfarin, phenprocoumon and acenocoumarol. *Clin Pharmacokinet* 2005;**44**:1227-1246.
17. Olie RH, Winckers K, Rocca B, Ten Cate H. Oral Anticoagulants Beyond Warfarin. *Annu Rev Pharmacol Toxicol* 2024;**64**:551-575.
18. Eliquis: EPAR – Product information. European Medicines Agency, 2025.
19. Xarelto: EPAR – Product information. European Medicines Agency, 2024.
20. Pradaxa : EPAR - Product Information. European Medicines Agency, 2024.
21. Lixiana : EPAR - Product Information. European Medicines Agency, 2024.
22. Stangier J. Clinical pharmacokinetics and pharmacodynamics of the oral direct thrombin inhibitor dabigatran etexilate. *Clin Pharmacokinet* 2008;**47**:285-295.
23. Hindley B, Lip GYH, McCloskey AP, Penson PE. Pharmacokinetics and pharmacodynamics of direct oral anticoagulants. *Expert Opin Drug Metab Toxicol* 2023;**19**:911-923.
24. Lin JH, Yamazaki M. Role of P-glycoprotein in pharmacokinetics: clinical implications. *Clin Pharmacokinet* 2003;**42**:59-98.
25. Fromm MF. Importance of P-glycoprotein at blood-tissue barriers. *Trends Pharmacol Sci* 2004;**25**:423-429.
26. Chan N, Sobieraj-Teague M, Eikelboom JW. Direct oral anticoagulants: evidence and unresolved issues. *Lancet* 2020;**396**:1767-1776.
27. Maliepaard M, Scheffer GL, Faneyte IF, van Gastelen MtA, Pijnenborg ACLM, Schinkel AH, van de Vijver MJ, Scheper RJ, Schellens JHM. Subcellular Localization

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and Distribution of the Breast Cancer Resistance Protein Transporter in Normal Human Tissues1. *Cancer Research* 2001;**61**:3458-3464.

28. Cooray HC, Blackmore CG, Maskell L, Barrand MA. Localisation of breast cancer resistance protein in microvessel endothelium of human brain. *Neuroreport* 2002;**13**:2059-2063.

29. Douxfils J, Adcock DM, Bates SM, Favaloro EJ, Gouin-Thibault I, Guillermo C, Kawai Y, Lindhoff-Last E, Kitchen S, Gosselin RC. 2021 Update of the International Council for Standardization in Haematology Recommendations for Laboratory Measurement of Direct Oral Anticoagulants. *Thromb Haemost* 2021;**121**:1008-1020.

30. Douxfils J, Ageno W, Samama CM, Lessire S, Ten Cate H, Verhamme P, Dogné JM, Mullier F. Laboratory testing in patients treated with direct oral anticoagulants: a practical guide for clinicians. *J Thromb Haemost* 2018;**16**:209-219.

31. Mbroh J, Birschmann I, Ebner M, Kremer Hovinga JA, Lindhoff-Last E, Purrucker J, Schäfer ST, Tünnerhoff J, Wang Y, Poli S. Rapid assessment of direct oral anticoagulants in acute stroke-An educational systematic review. *Eur Stroke J* 2025;**10**:24-34.

32. Godino C, Mazza R, Gaspardone C, Minerva A, Sena R, Cozzani G, Salerno A, Cera M, Slavich M, De Lorenzo LA, Leo G, Nemola G, Fattorini A, Crippa L, Della Valle P, Margonato A, D'Angelo A. Plasma levels measurement of the 4 direct oral anticoagulants in patients with atrial fibrillation at the time of acute thromboembolic and bleeding events. *J Thromb Haemost* 2025.

33. Douros A, Schlemm L, Bolbrinker J, Ebinger M, Kreutz R. Insufficient anticoagulation with dabigatran in a patient with short bowel syndrome. *Thromb Haemost* 2014;**112**:419-420.

34. Connolly SJ, Sharma M, Cohen AT, Demchuk AM, Członkowska A, Lindgren AG, Molina CA, Bereczki D, Toni D, Seiffge DJ, Tanne D, Sandset EC, Tsivgoulis G, Christensen H, Beyer-Westendorf J, Coutinho JM, Crowther M, Verhamme P, Amarenco P, Roine RO, Mikulik R, Lemmens R, Veltkamp R, Middeldorp S, Robinson TG, Milling TJ, Jr., Tedim-Cruz V, Lang W, Himmelmann A, Ladenvall P, Knutsson M, Ekholm E, Law A, Taylor A, Karyakina T, Xu L, Tsiplova K, Poli S, Kallmünzer B, Gumbinger C, Shoamanesh A. Andexanet for Factor Xa Inhibitor-Associated Acute Intracerebral Hemorrhage. *N Engl J Med* 2024;**390**:1745-1755.

35. Heo YA. Andexanet alfa in the treatment of acute major bleeding related to apixaban and rivaroxaban: a profile of its use in the USA. *Drugs Ther Perspect* 2018;**34**:507-512.

36. Almutairi AR, Zhou L, Gellad WF, Lee JK, Slack MK, Martin JR, Lo-Ciganic WH. Effectiveness and Safety of Non-vitamin K Antagonist Oral Anticoagulants for Atrial Fibrillation and Venous Thromboembolism: A Systematic Review and Meta-analyses. *Clin Ther* 2017;**39**:1456-1478.e1436.

37. Joosten LPT, van Doorn S, van de Ven PM, Köhlen BTG, Nierman MC, Koek HL, Hemels MEW, Huisman MV, Kruip M, Faber LM, Wiersma NM, Buding WF, Fijnheer R, Adriaansen HJ, Roes KC, Hoes AW, Rutten FH, Geersing GJ. Safety of Switching From a Vitamin K Antagonist to a Non-Vitamin K Antagonist Oral Anticoagulant in

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Frail Older Patients With Atrial Fibrillation: Results of the FRAIL-AF Randomized Controlled Trial. *Circulation* 2024;**149**:279-289.

38. Van Gelder IC, Rienstra M, Bunting KV, Casado-Arroyo R, Caso V, Crijns H, De Potter TJR, Dwight J, Guasti L, Hanke T, Jaarsma T, Lettino M, Løchen ML, Lumbers RT, Maesen B, Mølgaard I, Rosano GMC, Sanders P, Schnabel RB, Suwalski P, Svennberg E, Tamargo J, Tica O, Traykov V, Tzeis S, Kotecha D. 2024 ESC Guidelines for the management of atrial fibrillation developed in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS). *Eur Heart J* 2024;**45**:3314-3414.
39. Capodanno D, Alexander JH, Bahit MC, Eikelboom JW, Gibson CM, Goodman SG, Kunadian V, Lip GYH, Lopes RD, Mehran R, Mehta SR, Patel MR, Piccini JP, Rao SV, Ruff CT, Steg PG, Weitz JI, Angiolillo DJ. Factor XI inhibitors for the prevention and treatment of venous and arterial thromboembolism. *Nat Rev Cardiol* 2025.
40. Greco A, Laudani C, Spagnolo M, Agnello F, Faro DC, Finocchiaro S, Legnazzi M, Mauro MS, Mazzone PM, Occhipinti G, Rochira C, Scalia L, Capodanno D. Pharmacology and Clinical Development of Factor XI Inhibitors. *Circulation* 2023;**147**:897-913.
41. Goodnough LT, Levy JH. The Judicious Use of Recombinant Factor VIIa. *Semin Thromb Hemost* 2016;**42**:125-132.
42. Metze M, Klöter T, Stöbe S, Rechenberger B, Siegemund R, Siegemund T, Laufs U, Petros S, Pfrepper C. Plasma levels do not predict thrombin generation in patients taking direct oral anticoagulants. *Int J Lab Hematol* 2021;**43**:1539-1548.
43. Hagii J, Tomita H, Metoki N, Tamai Y, Saito S, Shiroto H, Hitomi H, Kamada T, Seino S, Takahashi K, Sasaki S, Yasujima M, Okumura K. Effect of rivaroxaban on prothrombin fragment 1+2 compared with warfarin in patients with acute cardioembolic stroke: Insight from its serial measurement. *Thromb Res* 2016;**148**:9-14.
44. Kirchhof P, Ezekowitz MD, Purmah Y, Schiffer S, Meng IL, Camm AJ, Hohnloser SH, Schulz A, Wosnitza M, Cappato R. Effects of Rivaroxaban on Biomarkers of Coagulation and Inflammation: A Post Hoc Analysis of the X-VeRT Trial. *TH Open* 2020;**4**:e20-e32.
45. Shaw JR, Maier CL, Tanaka K, Rocca B, Ten Cate H, Kim P, Carrier M, Connors JM, Levy JH. Thrombin Generation and the Pharmacodynamics of Parenteral Anticoagulants. 2025.
46. Ząbczyk M, Natorska J, Malinowski KP, Undas A. Effect of enoxaparin on plasma fibrin clot properties and fibrin structure in patients with acute pulmonary embolism. *Vascul Pharmacol* 2020;**133-134**:106783.
47. Yeromonahos C, Marlu R, Polack B, Caton F. Antithrombin-independent effects of heparins on fibrin clot nanostructure. *Arterioscler Thromb Vasc Biol* 2012;**32**:1320-1324.
48. Blombäck M, He S, Bark N, Wallen HN, Elg M. Effects on fibrin network porosity of anticoagulants with different modes of action and reversal by activated coagulation factor concentrate. *Br J Haematol* 2011;**152**:758-765.

ACCEPTED MANUSCRIPT

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49. Ammollo CT, Semeraro F, Semeraro N, Colucci M. The contribution of anti-Xa and anti-IIa activities to the profibrinolytic activity of low-molecular-weight heparins. *Thromb Haemost* 2009;**101**:782-785.
50. Paszek E, Undas A. Prevention of unfavorable fibrin clots and thrombo-embolic manifestations in patients with cardiovascular disease. *Expert Rev Cardiovasc Ther* 2025:1-16.
51. Ammollo CT, Semeraro F, Incampo F, Semeraro N, Colucci M. Dabigatran enhances clot susceptibility to fibrinolysis by mechanisms dependent on and independent of thrombin-activatable fibrinolysis inhibitor. *J Thromb Haemost* 2010;**8**:790-798.
52. Varin R, Mirshahi S, Mirshahi P, Lu-Hong L, Kierzek G, Vigneau J-F, Marie J-P, Perzborn E, Mirshahi M, Soria C, Soria J. Improvement of Thrombolysis by Rivaroxaban, An Anti Xa Inhibitor. Potential Therapeutic Importance in Patients with Thrombosis. *Blood* 2008;**112**:3031-3031.
53. Carter RLR, Talbot K, Hur WS, Meixner SC, Van Der Gugten JG, Holmes DT, Côté HCF, Kastrup CJ, Smith TW, Lee AYY, Pryzdial ELG. Rivaroxaban and apixaban induce clotting factor Xa fibrinolytic activity. *J Thromb Haemost* 2018;**16**:2276-2288.
54. Ageno W, Gallus AS, Wittkowsky A, Crowther M, Hylek EM, Palareti G. Oral anticoagulant therapy: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest* 2012;**141**:e44S-e88S.
55. Vanassche T, Hirsh J, Eikelboom JW, Ginsberg JS. Organ-specific bleeding patterns of anticoagulant therapy: lessons from clinical trials. *Thromb Haemost* 2014;**112**:918-923.
56. Eikelboom JW, Wallentin L, Connolly SJ, Ezekowitz M, Healey JS, Oldgren J, Yang S, Alings M, Kaatz S, Hohnloser SH, Diener HC, Franzosi MG, Huber K, Reilly P, Varrone J, Yusuf S. Risk of bleeding with 2 doses of dabigatran compared with warfarin in older and younger patients with atrial fibrillation: an analysis of the randomized evaluation of long-term anticoagulant therapy (RE-LY) trial. *Circulation* 2011;**123**:2363-2372.
57. de Jong CMM, Blondon M, Ay C, Buchmuller A, Beyer-Westendorf J, Biechele J, Bertoletti L, Colombo G, Donadini MP, Hendriks SV, Jara-Palomares L, Nopp S, Ruiz-Artacho P, Stephan P, Tromeur C, Vanassche T, Westerweel PE, Klok FA. Incidence and impact of anticoagulation-associated abnormal menstrual bleeding in women after venous thromboembolism. *Blood* 2022;**140**:1764-1773.
58. Kalmanti L, Lindhoff-Last E. Bleeding Issues in Women Under Oral Anticoagulation. *Hamostaseologie* 2022;**42**:337-347.
59. Álvarez-Sabín J, Maisterra O, Santamarina E, Kase CS. Factors influencing haemorrhagic transformation in ischaemic stroke. *Lancet Neurol* 2013;**12**:689-705.
60. Siegbahn A, Lindbäck J, Hijazi Z, Åberg M, Alexander JH, Eikelboom JW, Lopes RD, Pol T, Oldgren J, Granger CB, Yusuf S, Wallentin L. Multiplex protein screening of biomarkers associated with major bleeding in patients with atrial fibrillation treated with oral anticoagulation. *J Thromb Haemost* 2021;**19**:2726-2737.
61. Falanga A, Leader A, Ambaglio C, Bagoly Z, Castaman G, Elalamy I, Lecumberri R, Niessner A, Pabinger I, Szmit S, Trinchero A, Ten Cate H, Rocca B. EHA Guidelines

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on Management of Antithrombotic Treatments in Thrombocytopenic Patients With Cancer. *Hemasphere* 2022;**6**:e750.
62. Greinacher A. CLINICAL PRACTICE. Heparin-Induced Thrombocytopenia. *N Engl J Med* 2015;**373**:252-261.
63. Morgan RL, Ashoorion V, Cuker A, Begum H, Ross S, Martinez N, Chong BH, Linkins LA, Warkentin TE, Wiercioch W, Nieuwlaat R, Schünemann H, Santesso N. Management of heparin-induced thrombocytopenia: systematic reviews and meta-analyses. *Blood Adv* 2020;**4**:5184-5193.
64. Warkentin TE, Greinacher A. Laboratory Testing for Heparin-Induced Thrombocytopenia and Vaccine-Induced Immune Thrombotic Thrombocytopenia Antibodies: A Narrative Review. *Semin Thromb Hemost* 2023;**49**:621-633.
65. Salzman EW, Rosenberg RD, Smith MH, Lindon JN, Favreau L. Effect of heparin and heparin fractions on platelet aggregation. *J Clin Invest* 1980;**65**:64-73.
66. Zucker MB. Heparin and platelet function. *Fed Proc* 1977;**36**:47-49.
67. Jiritano F, Serraino GF, Ten Cate H, Fina D, Matteucci M, Mastroroberto P, Lorusso R. Platelets and extra-corporeal membrane oxygenation in adult patients: a systematic review and meta-analysis. *Intensive Care Med* 2020;**46**:1154-1169.
68. Shapiro MJ, Weiss EJ, Faruqi TR, Coughlin SR. Protease-activated receptors 1 and 4 are shut off with distinct kinetics after activation by thrombin. *J Biol Chem* 2000;**275**:25216-25221.
69. Coughlin SR. Protease-activated receptors and platelet function. *Thromb Haemost* 1999;**82**:353-356.
70. Wong PC, Jiang X. Apixaban, a direct factor Xa inhibitor, inhibits tissue-factor induced human platelet aggregation in vitro: comparison with direct inhibitors of factor VIIa, XIa and thrombin. *Thromb Haemost* 2010;**104**:302-310.
71. Chylova M, Motovska Z, Fialova A, Stetkarova I, Peisker T, Kalvach P. The effect of warfarin administration on platelet aggregation. *Bratisl Lek Listy* 2021;**122**:320-324.
72. Kawahito K, Adachi H, Ino T. Platelet aggregation in patients taking anticoagulants after valvular surgery: evaluation by a laser light-scattering method. *Journal of Artificial Organs* 2002;**5**:0188-0192.
73. Helgason CM, Hoff JA, Kondos GT, Brace LD. Platelet aggregation in patients with atrial fibrillation taking aspirin or warfarin. *Stroke* 1993;**24**:1458-1461.
74. Mieszczak C, Winther K. DOES WARFARIN ENHANCE PLATELET ACTIVITY? *Thrombosis Research* 1996;**84**:285-287.
75. Schiffer S, Schwers S, Heitmeier S. The effect of rivaroxaban on biomarkers in blood and plasma: a review of preclinical and clinical evidence. *J Thromb Thrombolysis* 2023;**55**:449-463.
76. Ringwala SM, Dibattiste PM, Schneider DJ. Effects on platelet function of a direct acting antagonist of coagulation factor Xa. *J Thromb Thrombolysis* 2012;**34**:291-296.
77. Polzin A, Dannenberg L, Thienel M, Orban M, Wolff G, Hohlfeld T, Zeus T, Kelm M, Petzold T. Noncanonical Effects of Oral Thrombin and Factor Xa Inhibitors in Platelet Activation and Arterial Thrombosis. *Thromb Haemost* 2021;**121**:122-130.

ACCEPTED MANUSCRIPT

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78. Steppich B, Dobler F, Brendel LC, Hessling G, Braun SL, Steinsiek AL, Deisenhofer I, Hyseni A, Roest M, Ott I. Effect of the FXa inhibitors Rivaroxaban and Apixaban on platelet activation in patients with atrial fibrillation. *J Thromb Thrombolysis* 2017;**43**:490-497.
79. Zemer-Wassercug N, Haim M, Leshem-Lev D, Orvin KL, Vaduganathan M, Gutstein A, Kadmon E, Mager A, Kornowski R, Lev EI. The effect of dabigatran and rivaroxaban on platelet reactivity and inflammatory markers. *J Thromb Thrombolysis* 2015;**40**:340-346.
80. Chan MY, Lin M, Lucas J, Moseley A, Thompson JW, Cyr D, Ueda H, Kajikawa M, Ortel TL, Becker RC. Plasma proteomics of patients with non-valvular atrial fibrillation on chronic anti-coagulation with warfarin or a direct factor Xa inhibitor. *Thromb Haemost* 2012;**108**:1180-1191.
81. Miyazawa K, Pastori D, Hammerstingl C, Cappato R, Meng IL, Kramer F, Cohen A, Schulz A, Eickels MV, Lip GYH, Marin F. Left atrial thrombus resolution in non-valvular atrial fibrillation or flutter: biomarker substudy results from a prospective study with rivaroxaban (X-TRA). *Ann Med* 2018;**50**:511-518.
82. Weisshaar S, Litschauer B, Gouya G, Mayer P, Smerda L, Kapiotis S, Kyrle PA, Eichinger S, Wolzt M. Antithrombotic triple therapy and coagulation activation at the site of thrombus formation: a randomized trial in healthy subjects. *J Thromb Haemost* 2014;**12**:1850-1860.
83. Petzold T, Thienel M, Dannenberg L, Mourikis P, Helten C, Ayhan A, M'Pembele R, Achilles A, Trojovky K, Konsek D, Zhang Z, Regenauer R, Pircher J, Ehrlich A, Lüsebrink E, Nicolai L, Stocker TJ, Brandl R, Röschenthaler F, Strecker J, Saleh I, Spannagl M, Mayr CH, Schiller HB, Jung C, Gerdes N, Hoffmann T, Levkau B, Hohlfeld T, Zeus T, Schulz C, Kelm M, Polzin A. Rivaroxaban Reduces Arterial Thrombosis by Inhibition of FXa-Driven Platelet Activation via Protease Activated Receptor-1. *Circulation Research* 2020;**126**:486-500.
84. Dannenberg L, M'Pembele R, Mourikis P, Helten C, Zako S, Ahlbrecht S, Richter H, Zikeli D, Benkhoff M, Huhn-Wientgen R, Thienel M, Levkau B, Kelm M, Petzold T, Polzin A. Rivaroxaban reduces thromboxane induced platelet aggregation - the forgotten compass arm? *Platelets* 2021;**32**:1126-1128.
85. Polzin A, Benkhoff M, Thienel M, Barcik M, Mourikis P, Shchurovska K, Helten C, Ehreiser V, Zhe Z, von Wulffen F, Theiss A, Peri S, Cremer S, Ahlbrecht S, Zako S, Wildeis L, Al-Kassis G, Metzen D, Utz A, Hu H, Vornholz L, Pavic G, Lüsebrink E, Strecker J, Tiedt S, Cramer M, Gliem M, Ruck T, Meuth SG, Zeus T, Mayr C, Schiller HB, Simon L, Massberg S, Kelm M, Petzold T. Long-term FXa inhibition attenuates thromboinflammation after acute myocardial infarction and stroke by platelet proteome alteration. *J Thromb Haemost* 2025;**23**:668-683.
86. Connolly SJ, Ezekowitz MD, Yusuf S, Eikelboom J, Oldgren J, Parekh A, Pogue J, Reilly PA, Themeles E, Varrone J, Wang S, Alings M, Xavier D, Zhu J, Diaz R, Lewis BS, Darius H, Diener HC, Joyner CD, Wallentin L. Dabigatran versus warfarin in patients with atrial fibrillation. *N Engl J Med* 2009;**361**:1139-1151.
87. Lip GY, Lane DA. Does warfarin for stroke thromboprophylaxis protect against MI in atrial fibrillation patients? *Am J Med* 2010;**123**:785-789.

ACCEPTED MANUSCRIPT

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88. Wienen W, Stassen JM, Priepke H, Ries UJ, Hauel N. In-vitro profile and ex-vivo anticoagulant activity of the direct thrombin inhibitor dabigatran and its orally active prodrug, dabigatran etexilate. *Thromb Haemost* 2007;**98**:155-162.
89. Nakazaki M, Oka S, Magota H, Kiyose R, Onodera R, Ukai R, Kataoka-Sasaki Y, Sasaki M, Honmou O. Pharmacological Difference Between Platelet Aggregations in Cardioembolic Stroke Patients with Direct Oral Anticoagulants: A Pilot Study. *J Stroke Cerebrovasc Dis* 2022;**31**:106520.
90. Achilles A, Mohring A, Dannenberg L, Grandoch M, Hohlfeld T, Fischer JW, Levkau B, Kelm M, Zeus T, Polzin A. Dabigatran enhances platelet reactivity and platelet thrombin receptor expression in patients with atrial fibrillation. *J Thromb Haemost* 2017;**15**:473-476.
91. Christersson C, Johnell M, Siegbahn A. The influence of direct thrombin inhibitors on the formation of platelet-leukocyte aggregates and tissue factor expression. *Thromb Res* 2010;**126**:e327-333.
92. Oi K, Shimizu M, Natori T, Tsuda K, Yoshida M, Kamada A, Ishigaku Y, Narumi S, Oura K, Maeda T, Terayama Y. Influence of PAR-1 in patients with non-valvular atrial fibrillation: The antiplatelet effect of dabigatran. *Thromb Res* 2021;**201**:123-130.
93. Perzborn E, Heitmeier S, Buetehorn U, Laux V. Direct thrombin inhibitors, but not the direct factor Xa inhibitor rivaroxaban, increase tissue factor-induced hypercoagulability in vitro and in vivo. *J Thromb Haemost* 2014;**12**:1054-1065.
94. Eikelboom JW, Connolly SJ, Bosch J, Dagenais GR, Hart RG, Shestakovska O, Diaz R, Alings M, Lonn EM, Anand SS, Widimsky P, Hori M, Avezum A, Piegas LS, Branch KRH, Probstfield J, Bhatt DL, Zhu J, Liang Y, Maggioni AP, Lopez-Jaramillo P, O'Donnell M, Kakkar AK, Fox KAA, Parkhomenko AN, Ertl G, Störk S, Keltai M, Ryden L, Pogosova N, Dans AL, Lanas F, Commerford PJ, Torp-Pedersen C, Guzik TJ, Verhamme PB, Vinereanu D, Kim JH, Tonkin AM, Lewis BS, Felix C, Yusoff K, Steg PG, Metsarinne KP, Cook Bruns N, Misselwitz F, Chen E, Leong D, Yusuf S. Rivaroxaban with or without Aspirin in Stable Cardiovascular Disease. *N Engl J Med* 2017;**377**:1319-1330.
95. Bonaca MP, Bauersachs RM, Anand SS, Debus ES, Nehler MR, Patel MR, Fanelli F, Capell WH, Diao L, Jaeger N, Hess CN, Pap AF, Kittelson JM, Gudz I, Mátyás L, Krievins DK, Diaz R, Brodmann M, Muehlhofer E, Haskell LP, Berkowitz SD, Hiatt WR. Rivaroxaban in Peripheral Artery Disease after Revascularization. *New England Journal of Medicine* 2020;**382**:1994-2004.
96. Borst O, Münzer P, Alnaggar N, Geue S, Tegtmeyer R, Rath D, Droppa M, Seizer P, Heitmeier S, Heemskerk JWM, Jennings LK, Storey RF, Angiolillo DJ, Rocca B, Spronk H, Ten Cate H, Gawaz M, Geisler T. Inhibitory mechanisms of very low–dose rivaroxaban in non–ST-elevation myocardial infarction. *Blood Advances* 2018;**2**:715-730.
97. Weiss L, O'Doherty A, Uhrig W, Szklanna PB, Hong-Minh M, Wynne K, Blanco A, Zivny J, Lima Passos V, Kevane B, Murphy S, F NÁ, O'Donnell M, Maguire PB. Rivaroxaban, in combination with low-dose aspirin, is associated with a reduction in proinflammatory and prothrombotic circulating vesicle signatures in patients with cardiovascular disease. *J Thromb Haemost* 2025;**23**:531-545.

ACCEPTED MANUSCRIPT

Downloaded from https://academic.oup.com/cardiovascres/advance-article/doi/10.1093/cvr/cvag043/8494234 by guest on 15 March 2026

98. Weiss L, Keaney J, Szklanna PB, Prendiville T, Uhrig W, Wynne K, Kelliher S, Ewins K, Comer SP, Egan K, O'Rourke E, Moran E, Petrov G, Patel A, Lennon Á, Blanco A, Kevane B, Murphy S, F NÁ, Maguire PB. Nonvalvular atrial fibrillation patients anticoagulated with rivaroxaban compared with warfarin exhibit reduced circulating extracellular vesicles with attenuated pro-inflammatory protein signatures. *J Thromb Haemost* 2021;**19**:2583-2595.
99. Weiss L, Uhrig W, Kelliher S, Szklanna PB, Prendiville T, Comer SP, Edebiri O, Egan K, Lennon Á, Kevane B, Murphy S, F NÁ, Maguire PB. Proteomic analysis of extracellular vesicle cargoes mirror the cardioprotective effects of rivaroxaban in patients with venous thromboembolism. *Proteomics Clin Appl* 2024;**18**:e202300014.
100. Russo V, Fabiani D, Leonardi S, Attena E, D'Alterio G, Cotticelli C, Rago A, Sarpa S, Maione B, D'Onofrio A, Golino P, Nigro G. Dual Pathway Inhibition with Rivaroxaban and Aspirin Reduces Inflammatory Biomarkers in Atherosclerosis. *J Cardiovasc Pharmacol* 2023;**81**:129-133.
101. Rizzi A, Petrucci G, Sacco M, Viti L, Brioschi M, Banfi C, Zaccardi F, Lancellotti S, Simone G, Cristofaro R, Rocca B, Pitocco D. Effects of low-dose rivaroxaban combined with low-dose aspirin versus low-dose aspirin alone on in vivo platelet activation, endothelial function and inflammation in type 2 diabetes patients with stable atherosclerotic disease: the RivAsa randomized, crossover study. *Diabetes Res Clin Pract* 2025;**224**:112244.
102. Groh LA, Willems LH, Fintelman P, Reijnen M, El Messaoudi S, Warlé MC. Dual-Pathway Inhibition with Rivaroxaban and Low-Dose Aspirin Does Not Alter Immune Cell Responsiveness and Distribution in Patients with Coronary Artery Disease. *Cardiol Ther* 2024;**13**:233-242.
103. Martins GL, Duarte RCF, Vieira É LM, Rocha NP, Figueiredo EL, Silveira FR, Caiaffa JRS, Lanna RP, Carvalho MDG, Palotás A, Ferreira CN, Reis HJ. Comparison of Inflammatory Mediators in Patients With Atrial Fibrillation Using Warfarin or Rivaroxaban. *Front Cardiovasc Med* 2020;**7**:114.
104. Kikuchi S, Tsukahara K, Sakamaki K, Morita Y, Takamura T, Fukui K, Endo T, Shimizu M, Sawada R, Sugano T, Himeno H, Kobayashi S, Arakawa K, Mochida Y, Tsunematsu T, Shigemasa T, Okuda J, Ishikawa T, Kimura K, Tamura K. Comparison of anti-inflammatory effects of rivaroxaban vs. dabigatran in patients with non-valvular atrial fibrillation (RIVAL-AF study): multicenter randomized study. *Heart Vessels* 2019;**34**:1002-1013.
105. Wu KK, Thiagarajan P. Role of endothelium in thrombosis and hemostasis. *Annu Rev Med* 1996;**47**:315-331.
106. Vannucchi S, Pasquali F, Chiarugi V, Ruggiero M. Internalization and metabolism of endogenous heparin by cultured endothelial cells. *Biochem Biophys Res Commun* 1986;**140**:294-301.
107. Muñoz EM, Linhardt RJ. Heparin-binding domains in vascular biology. *Arterioscler Thromb Vasc Biol* 2004;**24**:1549-1557.
108. Tan MH. The lipoprotein lipase system: new understandings. *Can Med Assoc J* 1978;**118**:675-680.

ACCEPTED MANUSCRIPT

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109. Zhang Y, Guo S, Xu J. Multifunctional applications and research advances of low-molecular-weight heparin. *Front Pharmacol* 2025;**16**:1585762.
110. Joyce DE, Gelbert L, Ciaccia A, DeHoff B, Grinnell BW. Gene expression profile of antithrombotic protein c defines new mechanisms modulating inflammation and apoptosis. *J Biol Chem* 2001;**276**:11199-11203.
111. Griffin JH, Fernández JA, Gale AJ, Mosnier LO. Activated protein C. *J Thromb Haemost* 2007;**5 Suppl 1**:73-80.
112. Esmon CT. Targeting factor Xa and thrombin: impact on coagulation and beyond. *Thromb Haemost* 2014;**111**:625-633.
113. Bae JS, Yang L, Rezaie AR. Lipid raft localization regulates the cleavage specificity of protease activated receptor 1 in endothelial cells. *J Thromb Haemost* 2008;**6**:954-961.
114. Atzemian N, Kareli D, Ragia G, Manolopoulos VG. Distinct pleiotropic effects of direct oral anticoagulants on cultured endothelial cells: a comprehensive review. *Front Pharmacol* 2023;**14**:1244098.
115. Gorzelak-Pabis P, Broncel M, Wojdan K, Gajewski A, Chalubinski M, Gawrysiak M, Wozniak E. Rivaroxaban protects from the oxysterol-induced damage and inflammatory activation of the vascular endothelium. *Tissue Barriers* 2021;**9**:1956284.
116. Gorzelak-Pabiś P, Broncel M, Pawlos A, Wojdan K, Gajewski A, Chałubiński M, Woźniak E. Dabigatran: its protective effect against endothelial cell damage by oxysterol. *Biomed Pharmacother* 2022;**147**:112679.
117. Torramade-Moix S, Palomo M, Vera M, Jerez D, Moreno-Castaño AB, Zafar MU, Rovira J, Diekmann F, Garcia-Pagan JC, Escolar G, Cases A, Diaz-Ricart M. Apixaban Downregulates Endothelial Inflammatory and Prothrombotic Phenotype in an In Vitro Model of Endothelial Dysfunction in Uremia. *Cardiovasc Drugs Ther* 2021;**35**:521-532.
118. Puech C, Delavenne X, He Z, Forest V, Mismetti P, Perek N. Direct oral anticoagulants are associated with limited damage of endothelial cells of the blood-brain barrier mediated by the thrombin/PAR-1 pathway. *Brain Res* 2019;**1719**:57-63.
119. Ten Cate H, Guzik TJ, Eikelboom J, Spronk HMH. Pleiotropic actions of factor Xa inhibition in cardiovascular prevention: mechanistic insights and implications for anti-thrombotic treatment. *Cardiovasc Res* 2021;**117**:2030-2044.
120. Li Q, Yang XT, Wei W, Hu XP, Li XX, Xu M. Favorable effect of rivaroxaban against vascular dysfunction in diabetic mice by inhibiting NLRP3 inflammasome activation. *J Cell Physiol* 2022;**237**:3369-3380.
121. Ito Y, Maejima Y, Nakagama S, Shiheido-Watanabe Y, Tamura N, Sasano T. Rivaroxaban, a Direct Oral Factor Xa Inhibitor, Attenuates Atherosclerosis by Alleviating Factor Xa-PAR2-Mediated Autophagy Suppression. *JACC Basic Transl Sci* 2021;**6**:964-980.
122. Pistrosch F, Matschke JB, Schipp D, Schipp B, Henkel E, Weigmann I, Sradnick J, Bornstein SR, Birkenfeld AL, Hanefeld M. Rivaroxaban compared with low-dose aspirin in individuals with type 2 diabetes and high cardiovascular risk: a

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randomised trial to assess effects on endothelial function, platelet activation and vascular biomarkers. *Diabetologia* 2021;**64**:2701-2712.

123. Correale M, Leopizzi A, Mallardi A, Ranieri A, Suriano MP, D'Alessandro D, Tricarico L, Mazzeo P, Tucci S, Pastore G, Maulucci G, Di Biase M, Brunetti ND. Switch to direct anticoagulants and improved endothelial function in patients with chronic heart failure and atrial fibrillation. *Thromb Res* 2020;**195**:16-20.

124. Vu HH, Moellmer SA, McCarty OJT, Puy C. New mechanisms and therapeutic approaches to regulate vascular permeability in systemic inflammation. *Curr Opin Hematol* 2025;**32**:130-137.

125. Moaveni DK, Lynch EM, Luke C, Sood V, Upchurch GR, Wakefield TW, Henke PK. Vein wall re-endothelialization after deep vein thrombosis is improved with low-molecular-weight heparin. *J Vasc Surg* 2008;**47**:616-624.

126. Sood V, Luke C, Miller E, Mitsuya M, Upchurch GR, Jr., Wakefield TW, Myers DD, Henke PK. Vein wall remodeling after deep vein thrombosis: differential effects of low molecular weight heparin and doxycycline. *Ann Vasc Surg* 2010;**24**:233-241.

127. Myers DD, Jr., Henke PK, Bedard PW, Wrobleski SK, Kaila N, Shaw G, Meier TR, Hawley AE, Schaub RG, Wakefield TW. Treatment with an oral small molecule inhibitor of P selectin (PSI-697) decreases vein wall injury in a rat stenosis model of venous thrombosis. *J Vasc Surg* 2006;**44**:625-632.

128. Lipowsky HH, Lescanic A. Inhibition of inflammation induced shedding of the endothelial glycocalyx with low molecular weight heparin. *Microvasc Res* 2017;**112**:72-78.

129. Bouman AC, Cheung YW, Spronk HM, Schalkwijk CG, ten Cate H, ten Wolde M, ten Cate-Hoek AJ. Biomarkers for post thrombotic syndrome: a case-control study. *Thromb Res* 2014;**134**:369-375.

130. Siudut J, Grela M, Wypasek E, Plens K, Undas A. Reduced plasma fibrin clot permeability and susceptibility to lysis are associated with increased risk of postthrombotic syndrome. *J Thromb Haemost* 2016;**14**:784-793.

131. Prandoni P, Ageno W, Ciammaichella M, Mumoli N, Zanatta N, Imberti D, Visonà A, Bucherini E, Di Nisio M, Noventa F. The risk of post-thrombotic syndrome in patients with proximal deep vein thrombosis treated with the direct oral anticoagulants. *Intern Emerg Med* 2020;**15**:447-452.

132. Aarts A, Brandts L, Pavlicic M, Ten Cate H, Ten Cate-Hoek AJ. Apixaban vs. vitamin K antagonists in relation to the post-thrombotic syndrome; data from a prospective cohort study. *J Thromb Haemost* 2025:In press.

133. Elango K, Javaid A, Khetarpal BK, Ramalingam S, Kolandaivel KP, Gunasekaran K, Ahsan C. The Effects of Warfarin and Direct Oral Anticoagulants on Systemic Vascular Calcification: A Review. *Cells* 2021;**10**.

134. Peeters F, Dudink E, Kimenai DM, Weijs B, Altintas S, Heckman LIB, Mihl C, Schurgers LJ, Wildberger JE, Meex SJR, Kietselaer B, Crijns H. Vitamin K Antagonists, Non-Vitamin K Antagonist Oral Anticoagulants, and Vascular Calcification in Patients with Atrial Fibrillation. *TH Open* 2018;**2**:e391-e398.

ACCEPTED MANUSCRIPT

Downloaded from https://academic.oup.com/cardiovascres/advance-article/doi/10.1093/cvr/cvag043/8494234 by guest on 15 March 2026

135. Plank F, Beyer C, Friedrich G, Stühlinger M, Hintringer F, Dichtl W, Wildauer M, Feuchtner G. Influence of vitamin K antagonists and direct oral anticoagulation on coronary artery disease: A CTA analysis. *Int J Cardiol* 2018;**260**:11-15.
136. Hasific S, Øvrehus KA, Gerke O, Hallas J, Busk M, Lambrechtsen J, Urbonaviciene G, Sand NPR, Nielsen JS, Diederichsen L, Pedersen KB, Carter-Storch R, Ilangkovan N, Mickley H, Rasmussen LM, Lindholt JS, Diederichsen A. Extent of arterial calcification by conventional vitamin K antagonist treatment. *PLoS One* 2020;**15**:e0241450.
137. Wu T, Huang J, Wang X, Lian H, Guo R, Shi C. Association of Oral Anticoagulant Therapy with the Prevalence and Severity of Vascular Calcification among Patients with Atrial Fibrillation: A Cohort Study. *ACS Pharmacol Transl Sci* 2024;**7**:1262-1269.
138. Lee J, Nakanishi R, Li D, Shaikh K, Shekar C, Osawa K, Nezarat N, Jayawardena E, Blanco M, Chen M, Sieckert M, Nelson E, Billingsley D, Hamal S, Budoff MJ. Randomized trial of rivaroxaban versus warfarin in the evaluation of progression of coronary atherosclerosis. *Am Heart J* 2018;**206**:127-130.
139. Win TT, Nakanishi R, Osawa K, Li D, Susaria SS, Jayawardena E, Hamal S, Kim M, Broersen A, Kitslaar PH, Dailing C, Budoff MJ. Apixaban versus warfarin in evaluation of progression of atherosclerotic and calcified plaques (prospective randomized trial). *Am Heart J* 2019;**212**:129-133.
140. Stöhr R, Reinartz S, Dirrichs T, Witte K, Schuh A, Brandenburg V. Rivaroxaban versus vitamin K antagonist treatment on the progression of coronary calcification: the IRIVASC-trial. *Sci Rep* 2024;**14**:17605.
141. Posthuma JJ, Posma JJN, van Oerle R, Leenders P, van Gorp RH, Jaminon AMG, Mackman N, Heitmeier S, Schurgers LJ, Ten Cate H, Spronk HMH. Targeting Coagulation Factor Xa Promotes Regression of Advanced Atherosclerosis in Apolipoprotein-E Deficient Mice. *Sci Rep* 2019;**9**:3909.
142. Posma JJ, Grover SP, Hisada Y, Owens AP, 3rd, Antoniak S, Spronk HM, Mackman N. Roles of Coagulation Proteases and PARs (Protease-Activated Receptors) in Mouse Models of Inflammatory Diseases. *Arterioscler Thromb Vasc Biol* 2019;**39**:13-24.
143. Habibi A, Ruf W, Schurgers L. Protease-activated receptors in vascular smooth muscle cells: a bridge between thrombo-inflammation and vascular remodelling. *Cell Commun Signal* 2025;**23**:57.
144. Ruf W, Graf C. Coagulation signaling and cancer immunotherapy. *Thromb Res* 2020;**191 Suppl 1**:S106-s111.
145. Graf C, Wilgenbus P, Pagel S, Pott J, Marini F, Reyda S, Kitano M, Macher-Göppinger S, Weiler H, Ruf W. Myeloid cell-synthesized coagulation factor X dampens antitumor immunity. *Sci Immunol* 2019;**4**.
146. Alexander ET, Minton AR, Hayes CS, Goss A, Van Ryn J, Gilmour SK. Thrombin inhibition and cyclophosphamide synergistically block tumor progression and metastasis. *Cancer Biol Ther* 2015;**16**:1802-1811.
147. Zamorano-Leon JJ, Serna-Soto M, Moñux G, Freixer G, Zekri-Nechar K, Cabrero-Fernandez M, Segura A, Gonzalez-Cantalapiedra A, Serrano J, Farré AL. Factor Xa

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Inhibition by Rivaroxaban Modified Mitochondrial-Associated Proteins in Human Abdominal Aortic Aneurysms. *Ann Vasc Surg* 2020;**67**:482-489.
148. Deng Z, Xie H, Cheng W, Zhang M, Liu J, Huo Y, Liao Y, Cheng Y. Dabigatran ameliorates airway smooth muscle remodeling in asthma by modulating Yes-associated protein. *J Cell Mol Med* 2020;**24**:8179-8193.
149. Iannucci J, Johnson SL, Majchrzak M, Barlock BJ, Akhlaghi F, Seeram NP, Sen A, Grammas P. Short-term treatment with dabigatran alters protein expression patterns in a late-stage tau-based Alzheimer's disease mouse model. *Biochem Biophys Rep* 2020;**24**:100862.
150. Narita Y, Hamamura K, Kashiyama M, Utsumi S, Kakizoe Y, Kondo Y, Ishitsuka Y, Jono H, Irie T, Mukoyama M, Saito H, Kadowaki D, Hirata S, Kitamura K. Edoxaban Exerts Antioxidant Effects Through FXa Inhibition and Direct Radical-Scavenging Activity. *Int J Mol Sci* 2019;**20**.
151. Samiei F, Sajjadi H, Jamshidzadeh A, Seydi E, Pourahmad J. Contrasting Role of Concentration in Rivaroxaban Induced Toxicity and Oxidative Stress in Isolated Kidney Mitochondria. *Drug Res (Stuttg)* 2019;**69**:523-527.
152. Kreutz R, Deray G, Floege J, Gwechenberger M, Hahn K, Luft AR, Persson P, Axthelm C, Beer JH, Bergler-Klein J, Lellouche N, Taggeselle J, Coleman CI, Beyer-Westendorf J. Rivaroxaban vs Vitamin K Antagonist in Patients With Atrial Fibrillation and Advanced Chronic Kidney Disease. *JACC Adv* 2024;**3**:100813.
153. Teymen B, Öner ME, Erdağ Y. Dual-pathway inhibition in patients with chronic limb-threatening ischemia requiring reintervention for infrapopliteal occlusions. *Heart Vessels* 2024;**39**:771-777.
154. Wang Z, Sheng L, Gu H, Yang F, Xie H, Li M. Rivaroxaban and Aspirin in Drug-Coated Balloon Angioplasty for Femoropopliteal In-Stent Restenosis: A Retrospective Cohort Study. *Ann Vasc Surg* 2024;**108**:338-345.
155. Jang G-W, Lee JM, Choi SW, Kim J, Lee YS, Kim HO, Chung H, Woo JS, Kim JB, Kim W-S, Kim W. Vascular Protective Effects of New Oral Anticoagulants in Patients with Atrial Fibrillation. *Journal of Clinical Medicine* 2021;**10**:4332.
156. Ikari Y, Saito F, Kiyooka T, Nagaoka M, Kimura M, Furuki T, Tanaka S. Switching from Warfarin to rivaroxaban induces sufficiency of vitamin K and reduction of arterial stiffness in patients with atrial fibrillation. *Heart Vessels* 2020;**35**:1727-1733.
157. Junejo RT, Gupta D, Snowdon RL, Lip GYH, Fisher JP. Relationship of Warfarin and Apixaban with Vascular Function in Patients with Atrial Fibrillation. *J Vasc Res* 2024;**61**:59-67.
158. Ngo ATP, Jordan KR, Mueller PA, Hagen MW, Reitsma SE, Puy C, Revenko AS, Lorentz CU, Tucker EI, Cheng Q, Hinds MT, Fazio S, Monia BP, Gailani D, Gruber A, Tavori H, McCarty OJT. Pharmacological targeting of coagulation factor XI mitigates the development of experimental atherosclerosis in low-density lipoprotein receptor-deficient mice. *J Thromb Haemost* 2021;**19**:1001-1017.
159. Mohammed BM, Cheng Q, Matafonov A, Verhamme IM, Emsley J, McCrae KR, McCarty OJT, Gruber A, Gailani D. A non-circulating pool of factor XI associated with glycosaminoglycans in mice. *J Thromb Haemost* 2019;**17**:1449-1460.

ACCEPTED MANUSCRIPT

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160. Pallares Robles A, Ten Cate V, Schulz A, Prochaska JH, Rapp S, Koeck T, Panova-Noeva M, Heitmeier S, Schwers S, Leineweber K, Seyfarth HJ, Opitz CF, Spronk H, Espinola-Klein C, Lackner KJ, Münzel T, Andrade-Navarro MA, Konstantinides SV, Ten Cate H, Wild PS. Association of FXI activity with thrombo-inflammation, extracellular matrix, lipid metabolism and apoptosis in venous thrombosis. *Sci Rep* 2022;**12**:9761.
161. van Es N, Coppens M, Schulman S, Middeldorp S, Büller HR. Direct oral anticoagulants compared with vitamin K antagonists for acute venous thromboembolism: evidence from phase 3 trials. *Blood* 2014;**124**:1968-1975.
162. Eikelboom JW, Weitz JI. Warfarin faring better: vitamin K antagonists beat rivaroxaban and apixaban in the INVICTUS and PROACT Xa trials. *J Thromb Haemost* 2023;**21**:3067-3071.
163. Bellido-Martín L, de Frutos PG. Vitamin K-dependent actions of Gas6. *Vitam Horm* 2008;**78**:185-209.
164. McShane L, Tabas I, Lemke G, Kurowska-Stolarska M, Maffia P. TAM receptors in cardiovascular disease. *Cardiovasc Res* 2019;**115**:1286-1295.
165. Zhao YF, Xu DC, Zhu GF, Zhu MY, Tang K, Li WM, Xu YW. Growth Arrest-Specific 6 Exacerbates Pressure Overload-Induced Cardiac Hypertrophy. *Hypertension* 2016;**67**:118-129.
166. Ma T, Huang R, Xu Y, Lv Y, Liu Y, Pan X, Dong J, Gao D, Wang Z, Zhang F, Yan C, Ong SB, Su Y, Xu D. Plasma GAS6 predicts mortality risk in acute heart failure patients: insights from the DRAGON-HF trial. *J Transl Med* 2023;**21**:21.
167. D'Alessandro E, Scaf B, Munts C, van Hunnik A, Trevelyan CJ, Verheule S, Spronk HMH, Turner NA, Ten Cate H, Schotten U, van Nieuwenhoven FA. Coagulation Factor Xa Induces Proinflammatory Responses in Cardiac Fibroblasts via Activation of Protease-Activated Receptor-1. *Cells* 2021;**10**.
168. Loubele ST, ten Cate H, Spronk HM. Anticoagulant therapy in critical organ ischaemia/reperfusion injury. *Thromb Haemost* 2010;**104**:136-142.
169. Gadi I, Fatima S, Elwakiel A, Nazir S, Mohanad Al-Dabet M, Rana R, Bock F, Manoharan J, Gupta D, Biemann R, Nieswandt B, Braun-Dullaeus R, Besler C, Scholz M, Geffers R, Griffin JH, Esmon CT, Kohli S, Isermann B, Shahzad K. Different DOACs Control Inflammation in Cardiac Ischemia-Reperfusion Differently. *Circ Res* 2021;**128**:513-529.
170. Cao Y, Wang Y, Zhou Z, Pan C, Jiang L, Zhou Z, Meng Y, Charugundla S, Li T, Allayee H, Seldin MM, Lusis AJ. Liver-heart cross-talk mediated by coagulation factor XI protects against heart failure. *Science* 2022;**377**:1399-1406.
171. Ji Y, Zhang MJ, Wang W, Norby FL, Eaton AA, Inciardi RM, Alonso A, Sedaghat S, Ganz P, Van't Hof J, Solomon SD, Chaves PHM, Heckbert SR, Shah AM, Chen LY. Association of Coagulation Factor XI Level With Cardiovascular Events and Cardiac Function in Community-Dwelling Adults: From ARIC and CHS. *Circulation* 2025;**151**:356-367.
172. Kossmann S, Lagrange J, Jäckel S, Jurk K, Ehlken M, Schönfelder T, Weihert Y, Knorr M, Brandt M, Xia N, Li H, Daiber A, Oelze M, Reinhardt C, Lackner K, Gruber A, Monia B, Karbach SH, Walter U, Ruggeri ZM, Renné T, Ruf W, Münzel T, Wenzel P. Platelet-

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localized FXI promotes a vascular coagulation-inflammatory circuit in arterial hypertension. *Sci Transl Med* 2017;**9**.

173. Yu X, Li Q, Zhang D, Guo K, Sun W, Huang W, Gao L, Yan R, Yun L, Wan J, Xu T, Wang P. KN060-a humanized dual-domain antibody simultaneously targeting FXI-A2/A3 epitopes: Pioneering the anti-coagulation factor XI strategy for Aterial hypertension - Evidence from preclinical models. *Eur J Pharmacol* 2025;**1006**:178171.

174. HEPARIN SODIUM, for intravenous use - accessdata.fda.gov. Food and Drug Administration, 2017.

175. Arixtra : EPAR - Product information. European Medicines Agency, 2008.

176. McKeage K, Plosker GL. Argatroban. *Drugs* 2001;**61**:515-522; discussion 523-514.

177. Beurskens DMH, Huckriede JP, Schrijver R, Hemker HC, Reutelingsperger CP, Nicolaes GAF. The Anticoagulant and Nonanticoagulant Properties of Heparin. *Thromb Haemost* 2020;**120**:1371-1383.

178. Fareed J, Jeske WP. Small-molecule direct antithrombins: argatroban. *Best Pract Res Clin Haematol* 2004;**17**:127-138.

179. Informatorium Medicamentorum. Koninklijke Nederlandse Maatschappij ter bevordering der Pharmacie.

180. Ruiz-Irastorza G, Tektonidou MG, Khamashta M. Anticoagulant and non-anticoagulant therapy in thrombotic antiphospholipid syndrome: old drugs and new treatment targets. *Rheumatology (Oxford)* 2024;**63**:Si96-si106.

181. Joglar JA, Chung MK, Armbruster AL, Benjamin EJ, Chyou JY, Cronin EM, Deswal A, Eckhardt LL, Goldberger ZD, Gopinathannair R, Gorenek B, Hess PL, Hlatky M, Hogan G, Ibeh C, Indik JH, Kido K, Kusumoto F, Link MS, Linta KT, Marcus GM, McCarthy PM, Patel N, Patton KK, Perez MV, Piccini JP, Russo AM, Sanders P, Streur MM, Thomas KL, Times S, Tisdale JE, Valente AM, Van Wagoner DR. 2023 ACC/AHA/ACCP/HRS Guideline for the Diagnosis and Management of Atrial Fibrillation: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation* 2024;**149**:e1-e156.

182. Key NS, Khorana AA, Kuderer NM, Bohlke K, Lee AYY, Arcelus JI, Wong SL, Balaban EP, Flowers CR, Gates LE, Kakkar AK, Tempero MA, Gupta S, Lyman GH, Falanga A. Venous Thromboembolism Prophylaxis and Treatment in Patients With Cancer: ASCO Guideline Update. *Journal of Clinical Oncology* 2023;**41**:3063-3071.

ACCEPTED MANUSCRIPT

Downloaded from https://academic.oup.com/cardiovascres/advance-article/doi/10.1093/cvr/cvag043/8494234 by guest on 15 March 2026

183. Cuker A, Arepally GM, Chong BH, Cines DB, Greinacher A, Gruel Y, Linkins LA, Rodner SB, Selleng S, Warkentin TE, Wex A, Mustafa RA, Morgan RL, Santesso N. American Society of Hematology 2018 guidelines for management of venous thromboembolism: heparin-induced thrombocytopenia. *Blood Adv* 2018;**2**:3360-3392.

184. Levine GN, McEvoy JW, Fang JC, Ibeh C, McCarthy CP, Misra A, Shah ZI, Shenoy C, Spinler SA, Vallurupalli S, Lip GYH, Cardiology obotAHACoC, Cardiovascular Co, Nursing S, Council S. Management of Patients at Risk for and With Left Ventricular Thrombus: A Scientific Statement From the American Heart Association. *Circulation* 2022;**146**:e205-e223.

185. Otto CM, Nishimura RA, Bonow RO, Carabello BA, Erwin JP, 3rd, Gentile F, Jneid H, Krieger EV, Mack M, McLeod C, O'Gara PT, Rigolin VH, Sundt TM, 3rd, Thompson A, Toly C. 2020 ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease: Executive Summary: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation* 2021;**143**:e35-e71.

186. Davis JPE, Lim JK, Francis FF, Ahn J. AGA Clinical Practice Update on Management of Portal Vein Thrombosis in Patients With Cirrhosis: Expert Review. *Gastroenterology* 2025;**168**:396-404.e391.

187. Anderson DR, Morgano GP, Bennett C, Dentali F, Francis CW, Garcia DA, Kahn SR, Rahman M, Rajasekhar A, Rogers FB, Smythe MA, Tikkinen KAO, Yates AJ, Baldeh T, Balduzzi S, Brożek JL, Ikobaltzeta IE, Johal H, Neumann I, Wiercioch W, Yepes-Nuñez JJ, Schünemann HJ, Dahm P. American Society of Hematology 2019 guidelines for management of venous thromboembolism: prevention of venous thromboembolism in surgical hospitalized patients. *Blood Adv* 2019;**3**:3898-3944.

188. Regitz-Zagrosek V, Roos-Hesselink JW, Bauersachs J, Blomström-Lundqvist C, Cífková R, De Bonis M, Iung B, Johnson MR, Kintscher U, Kranke P, Lang IM, Morais J, Pieper PG, Presbitero P, Price S, Rosano GMC, Seeland U, Simoncini T, Swan L, Warnes CA. 2018 ESC Guidelines for the management of cardiovascular diseases during pregnancy. *Eur Heart J* 2018;**39**:3165-3241.

189. Stevens SM, Woller SC, Kreuziger LB, Bounameaux H, Doerschug K, Geersing GJ, Huisman MV, Kearon C, King CS, Knighton AJ, Lake E, Murin S, Vintch JRE, Wells PS, Moores LK. Antithrombotic Therapy for VTE Disease: Second Update of the CHEST Guideline and Expert Panel Report. *Chest* 2021;**160**:e545-e608.

ACCEPTED MANUSCRIPT

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190. Cella A, Marè A, Gigli GL, Zedde M, Valente M, Merlino G. A pharmacokinetic and pharmacodynamic evaluation of asundexian: a novel factor XIa inhibitor for stroke prevention. *Expert Opin Drug Metab Toxicol* 2024;**20**:1003-1011.

191. Marè A, Cella A, Tereshko Y, Toraldo F, Gigli GL, Valente M, Merlino G. Milvexian, a novel factor XIa inhibitor for stroke prevention: pharmacokinetic and pharmacodynamic evaluation. *Expert Opin Drug Metab Toxicol* 2024;**20**:873-880.

192. Chatrou ML, Winckers K, Hackeng TM, Reutelingsperger CP, Schurgers LJ. Vascular calcification: the price to pay for anticoagulation therapy with vitamin K-antagonists. *Blood Rev* 2012;**26**:155-166.

193. Jannati S, Patnaik R, Banerjee Y. Beyond Anticoagulation: A Comprehensive Review of Non-Vitamin K Oral Anticoagulants (NOACs) in Inflammation and Protease-Activated Receptor Signaling. *Int J Mol Sci* 2024;**25**.

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1 **Table 1**: Pharmacological properties of parenteral anticoagulants (references see Supplementary information)

	UFH[6, 174]	**LMWH**[6, 9]	**Fondaparinux**[4, 175]	**DTI**[6, 176] Argatroban (A) Bivalirudin (B)	**FXI inhibitor**[3] Antisense oligonucleotides (A) Monoclonal antibodies (M)
Target	Antithrombin	Antithrombin	Antithrombin	FIIa	FXI(a)
Prodrug	No	No	No	No	No
Route	Intravenous Subcutaneous	Subcutaneous	Subcutaneous	Intravenous	Intravenous (M) Subcutaneous (AM)
Daily doses	Variable, continuous Fixed, twice	Fixed, once/twice	Fixed, once	Variable, continuous	Fixed, weekly-monthly
Monitoring	aPTT Anti-FXa	Anti-FXa	Anti-FXa	aPTT	aPTT
Antidote	Protamine	Protamine (partial)	*	*	*
Target effects	Inhibition of FIIa and FXa	Inhibition of FXa and limited inhibition of FIIa	Inhibition of FXa	Inhibition of FIIa	Inhibition of intrinsic coagulation pathway
Unintended effects (other than cardiovascular)	Osteoporosis due to binding osteoblasts and osteoclasts Anti-inflammatory properties because of ligand interactions[177]	Lower osteoporosis risk compared with UFH Anti-inflammatory properties because of ligand interactions[177]	-	Downregulation of inflammatory cytokines (A)[178]	-
Bioavailability (%)	100 (intravenous) 10-40 (subcutaneous)[179]	90	100	100	100 (intravenous)
Peak serum concentration (h)	Immediate (intravenous)	3-5	2	-	0.1-6 (AM) 7-21 days (M subcutaneous)
Plasma protein binding (%)	High, variable	Less than UFH	97-98.6	54 (A) 0 (B)[179]	-
Volume of distribution	0.07 L/kg	Close to plasma volume	7-11 L	0.2 L/kg (A)	-
Half-life (h)	0.5-2.5 (intravenous)	3-6	17	0.5-1	11-121 (M) 2-6 weeks (AM)
Renal excretion (%)	Variable	Predominant	64-77	0 (A) 20 (B)	Minimal (M)

2 *Abbreviations*: UFH, unfractionated heparin; LMWH, low-molecular-weight heparin; DTI, direct thrombin inhibitor;
3 F, factor; aPTT, activated partial thromboplastin time; h, hour; L, Liter; kg, kilogram. -: not reported. *: no specific
4 antidote available.

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2 **Table 2**: Recommended anticoagulant(s) per indication (references see Supplementary information)

Indication	Recommended anticoagulant(s)	Rationale
Antiphospholipid syndrome	VKA	Best balance between efficacy and risks[180] LMWH in case of contraindication for VKA, but long-term use might be inconvenient[180] Increased risk of arterial thrombosis with DOAC compared with VKA[180]
Atrial fibrillation	DOAC or VKA	Similar efficacy with lower risk of (notably intracranial) bleeding with DOAC compared with VKA[181] VKA are indicated in moderate-severe mitral stenosis, where DOAC exhibit inferior outcomes[181] VKA in case of contra-indication for DOAC[181]
Cancer-associated thrombosis	Direct factor Xa inhibitor or LMWH	Reduced recurrence rate but increased nonmajor bleeding risk with direct factor Xa inhibitor compared with LMWH, especially in gastrointestinal and genitourinary malignancies[182] Improved efficacy over VKA[182]
Heparin-induced thrombocytopenia	Argatroban, bivalirudin, danaparoid, fondaparinux, or DOAC	Choice of non-heparin anticoagulant depends on drug factors (e.g. availability), patient factors, and experience[183] Argatroban or bivalirudin may be preferred in critically ill patients because of their short half-life[183] Fondaparinux, danaparoid and DOAC (rivaroxaban most studied) may be more convenient to administer in stable patients[183] VKA increase the risk of limb gangrene by reducing protein C levels[62]
Left ventricular thrombus	VKA, LMWH, or DOAC	DOAC seems to be a reasonable alternative to VKA, especially if managing VKA therapy is difficult[184]
Mechanical heart valves	VKA	Reduced thrombus risk due to prosthetic material and abnormal flow conditions[185] DOAC not approved[185]
Portal vein thrombosis in liver cirrhosis	VKA, LMWH, or DOAC	VKA, LMWH, and DOAC can be used in CTP class A and B cirrhosis, only LMWH can be used in CTP class C cirrhosis[186] INR monitoring for VKA is unreliable[186]
Post-operative thromboprophylaxis	LMWH, UFH, or DOAC	LMWH or UFH preferred after major general surgery[187] DOAC preferred after total hip or knee arthroplasty, otherwise LMWH[187]
Anticoagulation during pregnancy	LMWH	Lowest foetal risks[188] Risk of embryopathy with VKA, mainly in first trimester[188]

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		DOAC contra-indicated, no adequate studies available[188]
Venous thromboembolism	DOAC	Similar efficacy with lower risk of (notably intracranial) bleeding with DOAC compared with VKA[189]

Abbreviations: VKA, vitamin K antagonist; LMWH, low-molecular-weight heparin; DOAC, direct oral anticoagulant; CTP, Child-Turcotte-Pugh; INR, international normalised ratio; UFH, unfractionated heparin.

Table 3: Pharmacological properties of oral anticoagulants (references, see Supplementary information)

	VKA[13, 16] Acenocoumarol (A) Phenprocoumon (P) Warfarin (W)	**DOAC**[18-21] Apixaban (A) Dabigatran (D) Edoxaban (E) Rivaroxaban (R)	**FXI inhibitor**[190, 191] Small molecules Asundexian (A) Milvexian (M)
Target	Vitamin K epoxide reductase	FXa (AER) FIIa (D)	FXIa
Prodrug	No	No (AER) Yes (D)	No
Route	Oral	Oral	Oral
Daily doses	Variable, once	Fixed, once (ER) Fixed, twice (AD)	Fixed, once/twice
Monitoring	INR	Anti-FXa (AER) Plasma diluted thrombin time (D)	-
Antidote	Vitamin K PCC	Andexanet alfa (AR) Idarucizumab (D)	-
Target effects	Depletion of vitamin K-dependent clotting factors	Inhibition of thrombin generation (AER) Inhibition of fibrin clot formation (D)	Inhibition of intrinsic coagulation pathway
Unintended effects (other than cardiovascular)	Cartilage calcification due to inactivation of matrix Gla protein[192] Accelerated bone loss and osteoporosis due to dysfunction of osteocalcin[2]	Modulation of inflammation via PAR-signalling[193]	-
Bioavailability (%)	60 (A) 90 (P) 100 (W)	50 (A) 6.5 (D) 62 (E) 80-100 (R)	100 (A)
Peak serum concentration (h)	1-3 (A)[179] 48-72 (P) [179] 0.3-4 (W)	3-4 (A) 0.5-2 (D) 1-2 (E) 2-4 (R)	1-4
Plasma protein binding (%)	>98 (A) >99 (PW)	87 (A) 35 (D) 55 (E) 92-95 (R)	-
Volume of distribution	0.22-0.52 L/kg (A)	21 L (A)	70.5 L (A)

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	0.11-0.14 L/kg (P) 0.11-0.18 L/kg (W)	60-70 L (D) 107 L (E) 50 L (R)	347 L (M)
Half-life (h)	6.6 (A) 110-125 (P) 35-58 (W)	12 (A) 11 (D) 10-14 (E) 5-13 (R)	14-18 (A) 8-18 (M)
Renal excretion (%)	65 (A) 65 (P) 80 (W)	27 (A) 85 (D) 35 (E) 33 (R)	20 (A) 7-18 (M)

Abbreviations: VKA, vitamin K antagonist; DOAC, direct oral anticoagulant; F, factor; INR, international normalised ratio; PCC, prothrombin complex concentrate; PAR, protease-activated receptor; h, hour; L, liter; kg, kilogram. -: not reported or unknown.

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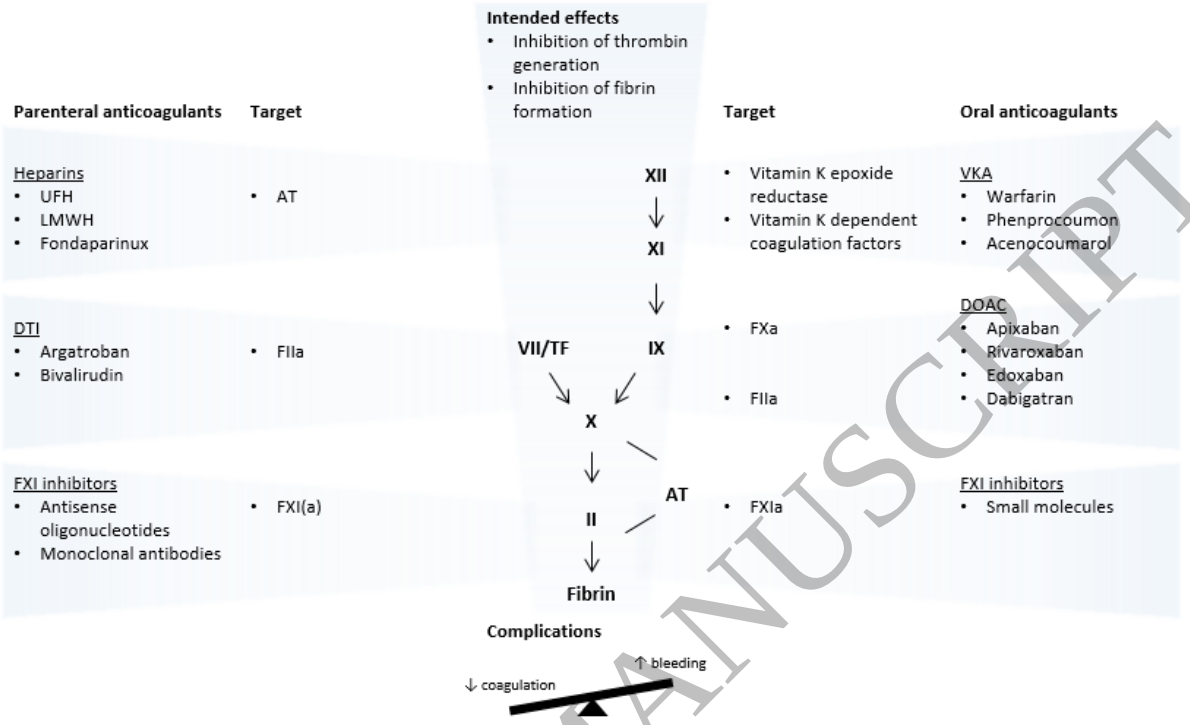

2

3 **Figure 1**: Intended effects of anticoagulants

4 *Abbreviations*: UFH, unfractionated heparin; LMWH, low-molecular-weight heparin; TF, tissue factor; AT,
5 antithrombin; DTI, direct thrombin inhibitor; VKA, vitamin K antagonist; DOAC, direct oral anticoagulant.

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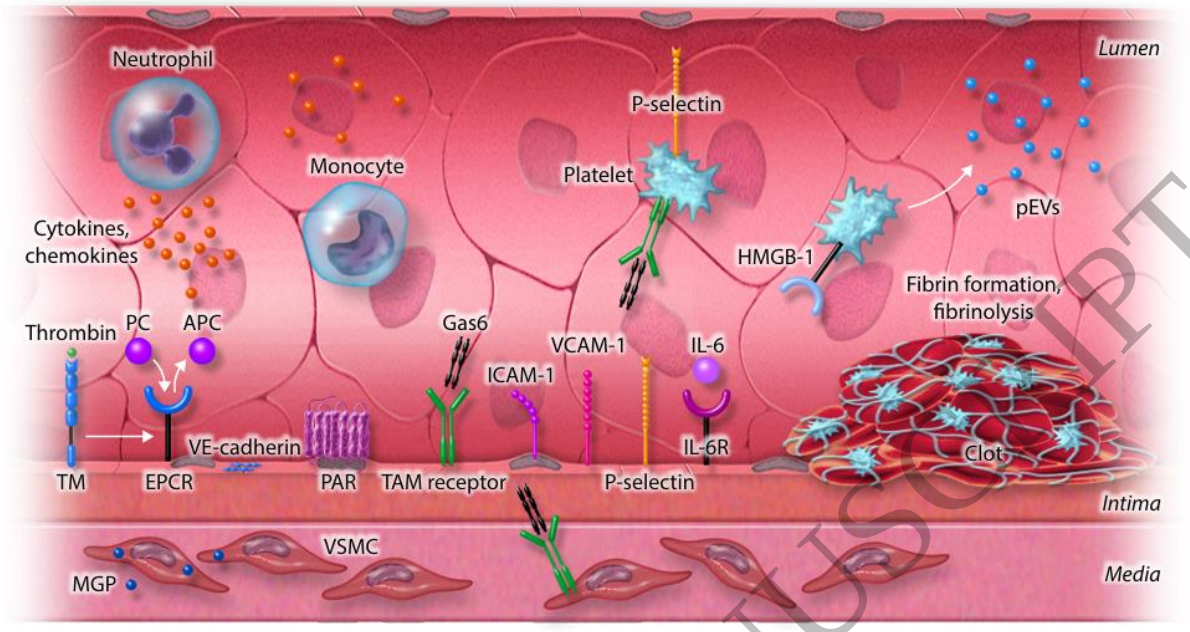

1

Target	Effect	Anticoagulant
The blood compartment		
Fibrin formation and lysis	↑ Clot permeability ↓ Lysis time	UFH, LMWH, VKA, DOAC
Platelet activation and aggregation	↑ Platelet activation ↓ Platelet aggregation ↓ P-selectin expression ↓ Thromboxane production ↓ HMGB-1 on platelets	VKA, dabigatran Dabigatran, rivaroxaban Rivaroxaban
EVs and immune modulation	↓ Platelet-derived EVs ↓ Inflammatory proteomic profile ↓ Pro-inflammatory cytokines ↑ Chemokines ↓ IL-6, fibrinogen	Rivaroxaban
Impact on endothelium	↑ APC (anti-inflammatory, prevents cell death) ↑ Restauration of endothelial barrier ↑ Expression of CAM ↓ Inflammation ↑ Anti-oxidant properties ↓ EC barrier function by cleavage of VE-cadherin	Rivaroxaban, dabigatran, apixaban However, dabigatran inhibited thrombin can still activate PAR-1 FXIa
Vessel wall and the heart		
Vessel wall	↑ Arterial media calcification ↑ PAR-1 activation (prevents calcification)	VKA (lesser DOAC) DOAC
Heart	↓ Synthesis of Gas6 (remodelling of cardiac and vascular tissues)	VKA

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	↓ FXIa associated with heart failure and fibrosis	FXI inhibitors

Figure 2: Unintended effects of anticoagulants

Abbreviations: UFH, unfractionated heparin; LMWH, low-molecular-weight heparin; VKA, vitamin K antagonist; DOAC, direct oral anticoagulant; HMGB-1, high mobility group box 1; EVs, extracellular vesicles; IL, interleukin; APC, activated protein C; CAM, cellular adhesion molecule; EC, endothelial cell; VE-cadherin, vascular endothelial cadherin; PAR, protease-activated receptor; Gas6, growth arrest-specific 6.

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